Appendix B

PROBABILITY AND STATISTICS FOR PCRA

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This Appendix contains a variety of probability and statistics results that are useful the context of portfolio construction and risk management theory and methods.

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B.1 Location and Scale Distributions

Let R be a random variable (think of R as a return, but it could be any random variable) with location parameter μ and scale parameter $s > 0$. The probability distribution function $F_R(r)$ and density function $f_R(r)$ of R have the forms

$$F_R(r) = F\left(\frac{r-\mu}{s}\right), \tag{B.1a}$$

$$f_R(r) = \frac{1}{s} f\left(\frac{r-\mu}{s}\right). \tag{B.1b}$$

Roughly speaking, the location parameter tells us where $f_R(r)$ is centered, while the scale parameter tells us how widely it spreads. The most famous member of the location–scale family of distributions is the Normal distribution, whose density function is

$$f_R(r) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(r-\mu)^2}{2\sigma^2}\right), \ -\infty < r < \infty. \tag{B.2}$$

The mapping from (B.2) to (B.1b) is obvious: the location parameter μ is the mean of the distribution, and the scale parameter σ is the square root of its variance σ^2. A closely related location-scale family is the family of t-distributions, with density function

$$f_R(r) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi\,\nu\,\tau^2}} \left(1 + \frac{1}{\nu}\,\frac{(r-\mu)^2}{\tau^2}\right)^{(\nu+1)/2}, \ -\infty < r < \infty. \tag{B.3}$$

location parameter μ, and scale parameter τ. For $\nu > 1$, the mean of the distribution is μ, and for $\nu > 2$, its variance is $\tau^2 \cdot \nu/\nu-2$. The mean of the t-distribution is undefined for $\nu = 1$, and its variance is undefined for $\nu \leqslant 2$. In the special case when $\nu = 1$, the t-distribution is known as a Cauchy distribution, and its mean is undefined, even though the distribution is centered at 0.

A less well-known, but still important, member of the location-scale family is the Laplacian, or two-sided exponential, distribution with density function

$$f_R(r) = \frac{1}{2b} \exp\left(-\frac{|r-\mu|}{b}\right), \ -\infty < r < \infty. \tag{B.4}$$

The location parameter μ is the mean of the distribution, and the scale parameter b is $1/\sqrt{2}$ times its standard deviation, or the square root of its variance. The Laplacian distribution can be shown to be a lateral translation by μ of the difference between two exponential random variables with pdf $\exp(-r/b)/b$[1].

[1] Most textbooks describe the exponential distribution in terms of a rate parameter λ that is the reciprocal of b. We deliberately use b in place of λ to emphasize the commonality with B.1b.

B.2 Invariance Properties of Location, Scale, Covariance and Correlation

Mean and Standard Deviation Equivariance

Let $\mu_R = \mathrm{E}[R]$ and $\sigma_R = \sqrt{\mathrm{var}(R)}$ be the mean and standard deviation of R, and let $\dot{R}$ be an *affine transformation*[2] of R defined as

$$\dot{R} = a + bR$$

where a is a location parameter and $b > 0$ is a scale parameter. It is straightforward to show that

$$\begin{aligned}\mu_{\dot{R}} &= a + b\mu_R \\ \sigma_{\dot{R}} &= b \cdot \sigma_R.\end{aligned}$$

The first equation above shows that the mean is *location and scale equivariant*, and the second equation shows that the standard deviation is *location invariant* and *scale equivariant*.

Covariance Translation Invariance and Scale Equivariance

Now suppose that r_i and r_j have covariance $\mathrm{cov}(r_i, r_j)$ and assume that variables $\dot{r}_i$ and $\dot{r}_j$ are obtained from r_i and r_j by the affine transformations

$$\begin{aligned}\dot{r}_i &= a + b \cdot r_i \\ \dot{r}_j &= c + d \cdot r_j.\end{aligned} \tag{B.5}$$

The resulting covariance, $\mathrm{cov}(\dot{r}_i, \dot{r}_j)$, of $\dot{r}_i$ and $\dot{r}_j$ is

$$\mathrm{cov}(\dot{r}_i, \dot{r}_j) = b \cdot d \cdot \mathrm{cov}(r_i, r_j), \tag{B.6}$$

and we describe this behavior by saying that covariances are *translation invariant* and *scale equivariant*.

Correlation Affine Invariance

Since the correlation of r_i and r_j is

$$\mathrm{cor}(r_i, r_j) = \frac{\mathrm{cov}(r_i, r_j)}{\sigma_{r_i} \cdot \sigma_{r_j}}, \tag{B.7}$$

it follows that follows that the correlation of $\dot{r}_i$ and $\dot{r}_j$ obtained by the affine transform (B.5) is the same as the correlation between r_i and r_j, namely

$$\mathrm{cor}(\dot{r}_i, \dot{r}_j) = \mathrm{cor}(r_i, r_j) \tag{B.8}$$

and we see that *correlations are affine invariant*.

[2] An affine transformation has both a constant and a linear term – it is best thought of as a parallel shift of a linear transformation.

Corresponding Properties of Sample Estimators

It is easy to check that the above equivariance and invariance properties also hold for the sample based estimators

B.3 Linearity of Covariances

Suppose we have two random variables, x and z, where $z = \sum_{i=1}^{n} a_i y_i$. Then

$$\begin{aligned}
\operatorname{cov}(x, z) &= \operatorname{cov}\left(x, \sum_{i=1}^{n} a_i y_i\right) \\
&= \mathrm{E}\left[(x - \mathrm{E}[x])\left(\sum_{i=1}^{n} a_i y_i - \sum_{i=1}^{n} a_i \mathrm{E}[y_i]\right)\right] \\
&= \mathrm{E}\left[\sum_{i=1}^{n} a_i (x - \mathrm{E}[x])(y_i - \mathrm{E}[y_i])\right] \\
&= \sum_{i=1}^{n} a_i \mathrm{E}[(x - \mathrm{E}[x])(y_i - \mathrm{E}[y_i])] \\
&= \sum_{i=1}^{n} a_i \operatorname{cov}(x, y_i) \\
&= \mathbf{a}' \operatorname{cov}(x, \mathbf{y}) \qquad \text{(B.9)}
\end{aligned}$$

where $\operatorname{cov}(x, \mathbf{y})$ is an n–dimensional column vector with i^{th} element $\operatorname{cov}(x, y_i)$.

A leading application of the above covariance linearity property is the computation of the covariance between the return of a portfolio, r_P, and its ith constituent asset, r_i. Clearly

$$r_P = \sum_{i=1}^{n} w_i r_i = \mathbf{w}'\mathbf{r}, \qquad \text{(B.10)}$$

and application of (B.9) gives:

$$\begin{aligned}
\operatorname{cov}(r_i, r_P) &= \sum_{j=1}^{n} w_j \operatorname{cov}(r_i, r_j) \\
&= \mathbf{w}' \operatorname{cov}(r_i, \mathbf{r}). \qquad \text{(B.11)}
\end{aligned}$$

B.4 Estimator Bias, Variance and Mean Squared Error

In this section we focus on estimators of a scalar parameter θ, whose value is unknown and is to be estimated using a set of random returns r_t, collected over n time periods $t = 1, 2, \ldots, n$. Using the vector notation $\mathbf{r} = (r_1, r_2, \cdots, r_n)'$, we denote the estimator as $\hat{\theta}_n(\mathbf{r}) = \hat{\theta}_n(r_1, r_2, \cdots, r_n)$. Since the estimator is a function of random returns, the estimator itself is a random variable whose basic properties are bias, variance, and mean squared error (MSE).

B.4.1 Unbiased versus Biased Estimators

An *unbiased* estimator $\hat{\theta}_n$ is one for which the expected value of the estimator is equal to the unknown true parameter value $\theta \in \mathbf{\Theta}$, where $\mathbf{\Theta}$ is a parameter space. For example, in the case of a location parameter, $\theta = \mu$, the parameter space will usually be the set of all finite real numbers, so that $\mathbf{\Theta} = \{x : -\infty < x < +\infty\}$. In the case of a variance parameter, $\theta = \sigma^2$, the parameter space will be the set all non–negative real numbers, $\mathbf{\Theta} = \{x : x \geqslant 0\}$, and if $\theta = \rho$ is a correlation coefficient, the parameter space is the set of all numbers less than or equal to one in magnitude, i.e. $\mathbf{\Theta} = \{x : |x| \leqslant 1\}$. Thus, the mathematical expression for unbiased of an estimator $\hat{\theta}_n$ is:

$$\mathrm{E}\left[\hat{\theta}_n\right] = \mathrm{E}\left[\hat{\theta}_n(\mathbf{r})\right] = \theta, \ \forall \theta \in \mathbf{\Theta}. \tag{B.12}$$

It is straightforward to show that the sample mean $\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t$ is an unbiased estimator of the true mean (expected value) μ.

When the above equality does not hold, the estimator is *biased*, and its bias is defined as:

$$B\left(\hat{\theta}_n\right) = \mathrm{E}\left[\hat{\theta}_n\right] - \theta, \forall \theta \in \mathbf{\Theta}. \tag{B.13}$$

As an example of a well–known, but slightly biased, estimator, consider the following sample variance estimator of an unknown true variance σ^2:

$$\hat{\sigma}_n^2 = \frac{1}{n}\sum_{t=1}^{n}\left(r_t - \bar{r}\right)^2. \tag{B.14}$$

It is known that for independent and identically distributed r_i with finite variance σ^2, the sample variance estimator

$$S_n^2 = \frac{1}{n-1}\sum_{t=1}^{n}\left(r_t - \bar{r}\right)^2 \tag{B.15}$$

is unbiased, i.e., $\mathrm{E}\left[S_n^2\right] = \sigma^2$. Thus the expected value of the sample variance (B.14) is

$$\mathrm{E}\left[\hat{\sigma}_n^2\right] = \mathrm{E}\left[\frac{1}{n}\sum_{t=1}^{n}\left(r_t - \bar{r}\right)^2\right] = \frac{n-1}{n}\mathrm{E}\left[S_n^2\right] = \frac{n-1}{n}\sigma^2 \tag{B.16}$$

and it has a negative bias

$$B\left(\hat{\sigma}_n^2\right) = \mathrm{E}\left[\hat{\sigma}_n^2\right] - \sigma^2 = -\frac{1}{n}\sigma^2 \tag{B.17}$$

that goes to zero as $n \to \infty$. Portfolio managers will seldom be concerned about such bias, which for a sample of size 50 is no larger than 2%.

Portfolio managers and investors generally pay little attention to returns variance, and instead focus on returns volatility, or returns standard deviation, and estimate it using the sample volatility estimator:

$$\hat{\sigma}_n = \sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(r_t - \bar{r}\right)^2}. \tag{B.18}$$

This volatility estimator is biased, as is the estimator with the divisor n replaced by $n-1$, even for normally distributed returns, but the bias can be shown to go to zero as the sample size tends to infinity.For normally distributed returns, Holtzman (1950) and Ahn and Fessler (2003) derive an unbiased estimate

of σ, and some experimentation shows that for $n \geqslant 4$, an almost perfectly unbiased estimate of σ can be obtained using the estimator

$$\hat{\sigma}_n = \sqrt{\frac{1}{n-1.5}\sum_{t=1}^{n}(r_t - \bar{r})^2}. \tag{B.19}$$

Principle of Unbiasedness

If possible, one should use an estimator that is unbiased for every finite sample size n. However, this often not possible, and one should then whenever possible use a *consistent* estimator, for which the finite sample bias goes to zero as the sample size tends to infinity. In the context of robust estimation, it may not be possible to obtain a consistent estimator, and this is discussed further in Section B.11.2.

B.4.2 Estimator Mean Squared Error

The mean squared error (MSE) of an estimator of a scalar quantity $\hat{\theta}_n = \hat{\theta}_n(r_1, r_2, \cdots, r_n)$ is defined as:

$$\mathrm{MSE}\left(\hat{\theta}_n\right) = \mathrm{E}\left[\left(\hat{\theta}_n - \theta\right)^2\right] \tag{B.20}$$

where we emphasize that the expected value depends implicitly on the distribution of the data $r_1, r_2, \cdots, r_n$, The right hand side of the above equation may be written as

$$\begin{aligned}\mathrm{E}\left[\left(\hat{\theta}_n - \theta\right)^2\right] &= \mathrm{E}\left[\left(\hat{\theta}_n - \mathrm{E}\left[\hat{\theta}_n\right] + \mathrm{E}\left[\hat{\theta}_n\right] - \theta\right)^2\right] \\ &= \mathrm{E}\left[\left(\hat{\theta}_n - \mathrm{E}\left[\hat{\theta}_n\right]\right)^2\right] + \mathrm{E}\left[\left(\mathrm{E}\left[\hat{\theta}_n\right] - \theta\right)^2\right] + \mathrm{E}\left[\left(\hat{\theta}_n - \mathrm{E}\left[\hat{\theta}_n\right]\right)\left(\mathrm{E}\left[\hat{\theta}_n\right] - \theta\right)\right].\end{aligned} \tag{B.21}$$

The first term above is the variance, $\mathrm{var}\left(\hat{\theta}_n\right)$, of the estimator, and the second term is the square of the bias $B\left(\hat{\theta}_n\right) = \mathrm{E}\left[\hat{\theta}_n\right] - \theta$. It is easy to show that the last term on the right hand side above is zero, giving us the variance plus squared bias decomposition of MSE:

$$\mathrm{MSE}\left(\hat{\theta}_n\right) = \mathrm{var}\left(\hat{\theta}_n\right) + B^2\left(\hat{\theta}_n\right). \tag{B.22}$$

There are three important things to remember about MSE:

1. For an unbiased estimator, the estimator MSE error is equal to its variance
2. For a biased estimator, one needs to be concerned about the squared bias contribution to MSE
3. Estimators can be applied to vector-valued quantities (e.g. a vector with the returns of all securities in the S&P 500) as well as to scalars: the MSE of an estimator $\hat{\mathbf{\Theta}}_n = \hat{\mathbf{\Theta}}_n(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_n)$ is the sum of the MSE's of its individual components, i.e.

$$\begin{aligned}\mathrm{MSE}\left(\hat{\boldsymbol{\Theta}}_n\right) &= \mathrm{E}\left[\left\|\hat{\boldsymbol{\Theta}}_n - \boldsymbol{\Theta}\right\|^2\right]. \\ &= \sum_i \mathrm{E}\left[\left(\hat{\boldsymbol{\Theta}}_{i,n} - \boldsymbol{\Theta}_i\right)^2\right] \\ &= \sum_i \mathrm{var}\left(\hat{\boldsymbol{\Theta}}_{i,n}\right) + B^2\left(\boldsymbol{\Theta}_i\right) \end{aligned} \tag{B.23}$$

Later, we will see that the Minimum MSE estimator is unbiased for a scalar parameter. The most commonly used estimators for a vector of parameters, including the sample mean, Ordinary Least Squares and sample covariance matrices, are also either unbiased or have biases of order $1/N$, where N is the number of parameters being estimated. But when estimating a vector of parameters, it is sometimes possible to admit a little bias in exchange for a much lower variance, so that the Minimum MSE estimator is slightly biased.

This remarkable result, due to Stein (1956), is counterintuitive: it is not easily proved or digested, and we refer the interested reader to the non-technical exposition in Efron and Morris (1977) as well as to the advanced treatment in Tsukuma and Kubokawa (2020). It also lies at the heart of asymptotic robust regression theory (see Section 5.8 of Maronna et al. (2019) as well as Yohai and Zamar (1997) and Svarc et al. (2002)), where the primary objective is to control the bias in parameter estimates under mixture models.

B.5 Least Squares

B.5.1 Ordinary Least Squares

We are often given a set of n observations $y_i,\ x_{i1}, x_{i2}, \cdots, x_{im},\ i = 1, 2, \cdots, n$, where y_i is a *response* variable (also known as a *dependent* variable) and the $x_{i1}, x_{i2}, \cdots, x_{im}$ are *predictor* variables (also called *independent* variables, or *explanatory* variables), and wish to model the relationship between the two using a *standard linear regression model*

$$y_i = \beta_0 + \sum_{j=1}^{m} \beta_j x_{ij} + \varepsilon_i,\ i = 1, 2, \cdots, n \tag{B.24}$$

where $\beta_0, \beta_1, \cdots, \beta_m$ are unknown *regression coefficients*. The first coefficient, β_0, is known as the *intercept*, and ε_i is an *error term*. The above model can be rewritten in the alternative *centered independent variables form*

$$y_i = \beta_0 + \sum_{j=1}^{m} \beta_j \bar{x}_j + \sum_{j=1}^{m} \beta_j \left(x_{ij} - \bar{x}_j\right) + \varepsilon_i,\ i = 1, 2, \cdots, n \tag{B.25}$$

where the $\bar{x}_j$ are the sample means, with respect to i, of the x_{ij}.

In many of the applications in this book, the response variable represents the returns of securities in a cross-section of size n, or in a time series at times $t = 1, 2, \cdots, n$, and the predictor variables are based on past and present financial statements, capital market data, and other sources of macro–economic information.

The *Ordinary Least Squares* (OLS) estimates $\hat{\boldsymbol{\beta}}_{OLS} = (\hat{\beta}_{0,OLS}, \hat{\beta}_{1,OLS}, \cdots, \hat{\beta}_{m,OLS})$ of the unknown coefficients $\beta_0, \beta_1, \cdots, \beta_m$ in (B.24) are the solution to the minimization problem

$$\min_{\beta_0,\, \beta_1,\, \cdots,\, \beta_m} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{m} \beta_j \, x_{ij} \right)^2 , \tag{B.26}$$

and the magnitude of the *residuals* $\hat{\varepsilon}_i = \hat{\varepsilon}_i(\hat{\boldsymbol{\beta}}_{OLS})$ reflect the model's goodness of fit.

The linear regression model and its OLS solution are conveniently described in matrix–vector notation by letting $\mathbf{y}$ be a column vector of response variables, $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m$ be column vectors of predictor variables, and $\boldsymbol{\epsilon}$ be a column vector of errors, respectively. The $n \times (m+1)$ matrix of predictor variables is

$$\mathbf{X} = [\mathbf{1}_n, \mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m] , \tag{B.27}$$

so that (B.24) can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{B.28}$$

The additional column of 1's in (B.27) corresponds to β_0 in (B.24), and captures the portion of the mean of $\mathbf{y}$ that is not accounted for by the means of $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m$, leaving the deviations of the $\mathbf{x}_i$ from their respective means to explain the deviations of $\mathbf{y}$ from its mean.

The OLS fit is the solution to[3]

$$\hat{\boldsymbol{\beta}}_{OLS} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \, (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' \, (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) , \tag{B.29}$$

and minimizes the sum of the squared errors

$$\boldsymbol{\epsilon} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}. \tag{B.30}$$

Expanding the right hand side of (B.29) and equating its gradient to $\mathbf{0}$ gives the well-known *normal equations*[4]

$$\begin{aligned} \mathbf{X}' \left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{OLS} \right) &= \mathbf{X}' \, \hat{\boldsymbol{\epsilon}} \\ &= \mathbf{0}. \end{aligned} \tag{B.31}$$

The OLS fit is a *projection* – it projects $\mathbf{y}$ onto the column space of $\mathbf{X}$, and its residuals are therefore orthogonal to *all* the columns of $\mathbf{X}$. In particular, they are orthogonal to $\mathbf{1}_n$, implying that the sample mean of the residuals, $\hat{\varepsilon}_i = \hat{\varepsilon}_i(\hat{\boldsymbol{\beta}}_{OLS})$, is 0, and that their sample second moment is equal to their sample variance with divisor n. The geometry of the OLS fit–plus–residuals decomposition is illustrated in Figure B.1.

[3] Hayashi (2000) has an excellent discussion of OLS, as do Weisberg (2014) and Greene (2018). Kevin Sheppard maintains a very useful set of notes for his financial econometrics class at Oxford University at `https://www.kevinsheppard.com/teaching/mfe/notes/` as does Bruce Hansen at `https://www.ssc.wisc.edu/~bhansen/econometrics/`.

[4] See Appendix A for the definitions of matrix multiplication and the gradient, as well as for formulae for the gradient of linear and quadratic forms.

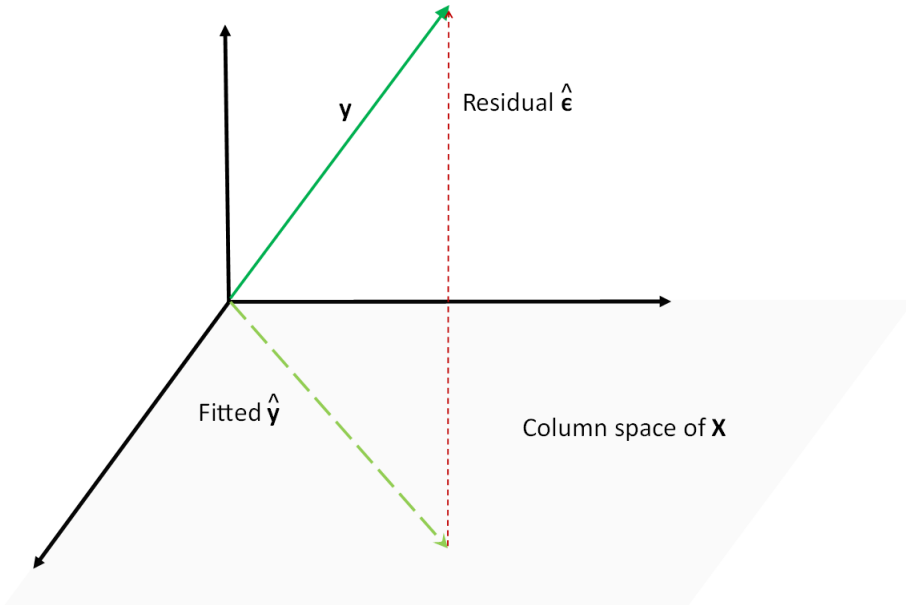

Fig. B.1 OLS Orthogonal Projection and the Fit–plus–Residuals Decomposition

If $\mathbf{X}$ has full rank, i.e., if its columns are linearly independent, then $\mathbf{X}'\mathbf{X}$ is invertible, and the left hand side of (B.31) is easily rearranged to give

$$\hat{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1}\,\mathbf{X}'\mathbf{y}, \tag{B.32}$$

and the corresponding decomposition of $\mathbf{y}$ into the sum of its OLS fit plus a residual is

$$\hat{\mathbf{y}} = \mathbf{X}\,\hat{\boldsymbol{\beta}}_{OLS} \tag{B.33a}$$

$$\mathbf{y} = \hat{\mathbf{y}} + \hat{\boldsymbol{\epsilon}}. \tag{B.33b}$$

If we subsitute (B.32) into (B.33a), we get

$$\begin{aligned}\hat{\mathbf{y}} &= \mathbf{X}\,(\mathbf{X}'\mathbf{X})^{-1}\,\mathbf{X}'\mathbf{y}\\ &= \mathbf{H}\mathbf{y},\end{aligned} \tag{B.34}$$

where

$$\mathbf{H} = \mathbf{X}\,(\mathbf{X}'\mathbf{X})^{-1}\,\mathbf{X}' \tag{B.35}$$

is variously known as the *Hat matrix* or the *Projection matrix* as it projects $\mathbf{y}$ onto $\hat{\mathbf{y}}$, which lies in the column space of $\mathbf{X}$. The hat matrix is symmetric and idempotent, i.e.

$$\mathbf{H}' = \mathbf{H} \tag{B.36a}$$

$$\mathbf{H} = \mathbf{H}^2. \tag{B.36b}$$

It is easily seen that (B.33b) can be rewritten using the Hat matrix as

$$\begin{aligned} \hat{\boldsymbol{\epsilon}} &= (\mathbf{I} - \mathbf{H})\,\mathbf{y}. \\ &= \mathbf{M}\,\mathbf{y}, \end{aligned} \tag{B.37}$$

where

$$\mathbf{M} = \mathbf{I} - \mathbf{H} \tag{B.38}$$

is known as the *Annihilator matrix*.

Even though (B.24) is linear in $\mathbf{x}$, it offers a powerful way in which to incorporate some, but not all, of the nonlinearity in the relationship between the response variable and the explanatory variables: for example, the columns of $\mathbf{X}$ can be non–linear transformations, and even cross–products that capture possibly important interactions, of the individual explanatory variables being used to predict $\mathbf{y}$. In many cases of practical interest, this simple modification significantly improves the fit generated by (B.33a) [5], while taking advantage of the computational efficiency of a linear model.

It is important to keep in mind that even though OLS is blessed with an elegant theoretical foundation, it is not at all robust to outliers: the OLS coefficients are linear in $\mathbf{y}$, and consequently, even a single outlier can make the bias in $\hat{\boldsymbol{\beta}}_{OLS}$ arbitrarily large[6]. This greatly reduces its utility in fields such as finance where data is plagued with outliers: OLS estimates can prove very misleading, and it is for this reason that we stress the use of robust statistical methods throughout this book.

B.5.2 Weighted Least Squares

Our development of OLS implicitly assumes that all observations are equally important. But this is not necessarily true – an investor estimating the beta of a stock, for example, may be much more concerned with its behavior in down markets than in up markets, or on days when the market makes large moves. Equations (B.29) and (B.32) are easily modified to accommodate this requirement by overweighting some observations and underweighting others.

Let $\mathbf{W}$ be a diagonal *weight matrix*, with $\mathbf{W}_{ii} \geqslant 0$ being the weight assigned to the ith observation. As $\mathbf{W}$ is diagonal, and as all of its entries are non–negative, it can be written as a product

$$\mathbf{W} = \mathbf{V}'\,\mathbf{V}, \tag{B.39}$$

where the diagonal entries of $\mathbf{V}$ and $\mathbf{W}$ are related by

[5] This technique is exploited in Chapter 12 when forecasting the returns of the S&P 500 using its earnings and revenues.

[6] Heitman and Ord (1985) illuminate OLS's lack of robustness in a different way – they show that $\hat{\boldsymbol{\beta}}_{OLS}$ is a weighted average of all $n!/(m+1)!(n-m-1)!$ possible coefficient vectors that can be formed using linear combinations of $m+1$ rows of $\mathbf{X}$ and the corresponding rows of $\mathbf{y}$. A change to any single row induces a change in each of the $(n-1)!/m!(n-m-1)!$ coefficient vectors formed using it, making the weighted average not robust to even a single outlier.

$$v_{ii} = \sqrt{w_{ii}},\ 1 \leqslant i \leqslant n. \tag{B.40}$$

To compute a down–market beta, for example, we set $v_{ii} = w_{ii} = 0$ for all i such that $r_{m,i} \geqslant 0$ and $v_{ii} = w_{ii} = 1$ for all i such that $r_{m,i} < 0$. The *Weighted Least Squares* (WLS) fit is the solution to

$$\hat{\boldsymbol{\beta}}_{WLS} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)' \mathbf{W} \left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right). \tag{B.41}$$

The normal equations now become

$$\begin{aligned} \mathbf{X}'\mathbf{W}\left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{WLS}\right) &= \mathbf{X}'\mathbf{W}\hat{\boldsymbol{\epsilon}} \\ &= \mathbf{0}, \end{aligned} \tag{B.42}$$

so that the residuals are orthogonal to the weighted regressors. Furthermore, if $\mathbf{X}'\mathbf{W}\mathbf{X}$ is invertible[7], the left hand side of (B.42) can be rearranged to give

$$\hat{\boldsymbol{\beta}}_{WLS} = \left(\mathbf{X}'\mathbf{W}\mathbf{X}\right)^{-1} \mathbf{X}'\mathbf{W}\mathbf{y}. \tag{B.43}$$

Equation (B.41) can be rewritten as

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{WLS} &= \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\mathbf{V}\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)\right)'\mathbf{V}\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right) \\ &= \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\left(\mathbf{V}\mathbf{y}\right) - \left(\mathbf{V}\mathbf{X}\right)\boldsymbol{\beta}\right)' \left(\left(\mathbf{V}\mathbf{y}\right) - \left(\mathbf{V}\mathbf{X}\right)\boldsymbol{\beta}\right). \end{aligned} \tag{B.44}$$

Equation (B.44) makes clear the fact that WLS is just an application of OLS to a scaled set of response and predictor variables. Alternatively, OLS can be viewed as a special case of (B.41) in which $\mathbf{W} = \mathbf{V} = \mathbf{I}_n$.

If the errors are heteroskedastic, i.e., if their variances are not all the same, then the weights can be chosen to reflect the uncertainty in the observations. A natural choice for the weights is $w_{ii} = 1/\sigma_i^2$, where σ_i^2 is the known variance of ε_i. In fact, if the errors are uncorrelated and normally distributed with mean 0, the WLS estimator with these weights is a maximum likelihood estimator. This can be seen by letting p_0 in (B.127) be a standard normal distribution, replacing s with σ_i, and then carrying out the indicated minimization.

In most cases of interest, the error variances are not known in advance, but must be estimated from the data, diminishing the benefits of WLS. Even so, the WLS estimate often represents a significant improvement over its OLS counterpart. Robust regression methods, in particular, are at heart specialized variants of WLS that automate the process of identifying and underweighting outliers using data dependent weights, which greatly diminishes their impact on the solution to (B.41). It is shown in Chapters 17 and 18 that data dependent weights can be represented as

$$w_{ii} = \frac{\psi\left(\hat{\varepsilon}_i\right)}{\hat{\varepsilon}_i}, \tag{B.45}$$

where $\psi\left(.\right)$ is a nonlinear transformation that is applied to the residuals, and which systematically underweights outliers.

[7] This is equivalent to the condition that $\mathbf{X}'\mathbf{V}$ has full rank.

B.5.3 Generalized Least Squares

Generalized Least Squares (GLS) combines two important ideas to identify an optimal set of coefficients for the linear regression model when its errors are correlated.

1. Individual observations can be weighted in inverse proportion to the variance of their residuals to equalize their relative importance (the same idea was earlier applied in our discussion of Weighted Least Squares), and
2. The distance between a vector of correlated residuals and the mean of their joint distribution can be computed using a scale invariant distance measure known as the *Mahalanobis distance*[8].

Formally, the GLS fit is the solution to

$$\hat{\boldsymbol{\beta}}_{GLS} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)' \mathbf{C}^{-1} \left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right), \tag{B.46}$$

where $\mathbf{C}$ is the known, non–singular, covariance matrix of the errors. Replacing our diagonal weight matrix $\mathbf{W}$ with $\mathbf{C}^{-1}$ simultaneously scales and decorrelates the errors, and equalizes the relative importance of each term in (B.46).

If $\mathbf{X}'\mathbf{C}^{-1}\mathbf{X}$ is invertible, the GLS coefficients are

$$\hat{\boldsymbol{\beta}}_{GLS} = \left(\mathbf{X}'\mathbf{C}^{-1}\mathbf{X}\right)^{-1} \mathbf{X}'\mathbf{C}^{-1}\mathbf{y}. \tag{B.47}$$

As $\mathbf{C}^{-1}$ is square, symmetric, and positive definite, it can be factored as[9]

$$\mathbf{C}^{-1} = \boldsymbol{\Omega}'\boldsymbol{\Omega}, \tag{B.48}$$

and we can rewrite (B.46) as

$$\hat{\boldsymbol{\beta}}_{GLS} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\left(\boldsymbol{\Omega}\mathbf{y}\right) - \left(\boldsymbol{\Omega}\mathbf{X}\right)\boldsymbol{\beta}\right)' \left(\left(\boldsymbol{\Omega}\mathbf{y}\right) - \left(\boldsymbol{\Omega}\mathbf{X}\right)\boldsymbol{\beta}\right), \tag{B.49}$$

illuminating the fact that GLS is just the application of OLS to a particular transformation of the raw data[10].

Unfortunately, $\hat{\boldsymbol{\beta}}_{GLS}$, like $\hat{\boldsymbol{\beta}}_{OLS}$, is linear in $\mathbf{y}$, and GLS therefore inherits OLS's primary failing – it is not robust to the presence of even a single outlier. In addition, $\mathbf{C}$ is usually not known, and must be estimated from data. The estimation process is inherently noisy, and widens the distribution of $\hat{\boldsymbol{\beta}}_{GLS}$, eroding much of the theoretical advantage of GLS over WLS and OLS. Furthermore, the finite sample properties of the GLS estimator with an unknown covariance matrix are not known, and as a consequence, OLS and WLS see far more use in practice than does GLS.

B.5.4 Least Squares Inference with Normally Distributed Errors

In some special cases, OLS yields information about the distribution of the regression coefficients, allowing us to test hypotheses about their value. In particular, if the errors have an i.i.d. normal distribu-

[8] See Appendix A for a discussion of the Mahalanobis distance.

[9] See the discussion of the Singular Value Decomposition in Appendix A,

[10] See Chapter 1 of Hayashi (2000) or Chapter 9 of Greene (2018) for a lucid description of GLS and its properties.

tion, i.e., if

$$\epsilon \,|\, \mathbf{X} \sim \text{MVN}\left(0,\ \sigma^2 \mathbf{I}_n\right)\ \forall \mathbf{X}, \tag{B.50}$$

the sampling distribution of the regression coefficients is also multivariate normal, as

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{OLS} - \boldsymbol{\beta} &= \left(\mathbf{X}'\mathbf{X}\right)^{-1} \mathbf{X}'\mathbf{y} - \boldsymbol{\beta} \\ &= \left(\mathbf{X}'\mathbf{X}\right)^{-1} \mathbf{X}' \left(\mathbf{X}\boldsymbol{\beta} + \epsilon\right) - \boldsymbol{\beta} \\ &= \left(\mathbf{X}'\mathbf{X}\right)^{-1} \mathbf{X}'\epsilon. \end{aligned} \tag{B.51}$$

It follows that the covariance matrix of the sampling errors of the regression coefficients is given by

$$Cov\left(\hat{\boldsymbol{\beta}}_{OLS} \,|\, \mathbf{X}\right) \sim \text{MVN}\left(0,\ \sigma^2 \left(\mathbf{X}'\mathbf{X}\right)^{-1}\right). \tag{B.52}$$

Even though σ^2 is unknown, we can obtain an unbiased estimate of it using the squared residuals [11]

$$s^2 = \frac{\hat{\epsilon}'\,\hat{\epsilon}}{n - m - 1}, \tag{B.53}$$

using which the standard error of the jth regression coefficient can be computed as:

$$SE\left(\hat{\beta}_{j,OLS}\right) = \sqrt{s^2 \times \left(\left(\mathbf{X}'\mathbf{X}\right)^{-1}\right)_{j,j}},\ 1 \leqslant j \leqslant m+1. \tag{B.54}$$

If we have a hypothesized value for β_j, say $\overline{\beta_j}$, we can test the hypothesis

$$H_0:\ \beta_j = \overline{\beta_j} \tag{B.55}$$

using the t–test, whose t–statistic

$$t_j = \frac{\hat{\beta}_{j,OLS} - \overline{\beta_j}}{SE\left(\hat{\beta}_{j,OLS}\right)}, \tag{B.56}$$

has a t distribution with $n - m - 1$ degrees of freedom.

If we specify a two–sided significance level α, we can reject the null hypothesis if the t–statistic is sufficiently large, i.e., if

$$|t_j| > t_{\alpha/2}\left(n - m - 1\right). \tag{B.57}$$

Often, $\overline{\beta_j}$ is set to 0, so that the null hypothesis corresponds to x_j being unrelated to y and thus superfluous to the model, and its rejection corresponds to the identification of a useful explanatory variable. The t–test finds use with non–zero values of $\overline{\beta_j}$ as well: for example, when testing the validity of the single–factor CAPM model for a young, fast–growing firm with a volatile stock, the null hypothesis might reflect our belief that its true beta is 2, and this hypothesis can be tested by running a regression followed by a t–test.

More generally, we can use the F test to test the linear hypothesis

$$H_0:\ \mathbf{A}\boldsymbol{\beta} = \mathbf{a}, \tag{B.58}$$

[11] The denominator of (B.53) is $n - m - 1$, and not $n - m$, because we have a total of $m + 1$ explanatory variables: an intercept and m predictor variables. In some textbooks, x_1 is assumed to be the intercept, which reduces the number of explanatory variables to m and makes the denominator $n - m$.

where $\mathbf{a}$ is an $p \times 1$ vector, and $\mathbf{A}$ is a $p \times (m+1)$ matrix with rank p. Common uses of this specification include testing the hypothesis that two or more of the coefficients are simultaneously equal to pre–specified values (often 0), or that their sum or difference is equal to a constant. In fact, (B.55) can be viewed as a special case of (B.58) in which $\mathbf{a} = 0$ and $\mathbf{A}$ is a $1 \times (m+1)$ matrix with 1 in its $j+1^{st}$ column and 0's elsewhere.

It is shown in Proposition 1.4 of Hayashi (2000) that under the null hypothesis (B.58), the F statistic

$$F = \frac{\left(\mathbf{A}\hat{\boldsymbol{\beta}}_{OLS} - \mathbf{a}\right)' \left(\mathbf{A}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{A}'\right)^{-1} \left(\mathbf{A}\hat{\boldsymbol{\beta}}_{OLS} - \mathbf{a}\right)}{p \cdot s^2} \tag{B.59}$$

with s^2 defined in (B.53), has a $F\left(p,\ n-m-1\right)$ distribution with p degrees of freedom in the numerator and $n-m-1$ degrees of freedom in the denominator.

The null hypothesis is rejected at significance level α if

$$F > F_{\alpha}\left(p,\ n-m-1\right), \tag{B.60}$$

where $F_{\alpha}\left(.\right)$ is the critical value of the F distribution corresponding to a tail probability of α. The t and F distributions are related: it can be shown that if a random variable u has a $t(q)$ distribution, u^2 has an $F\left(1,\ q\right)$ distribution.

The F test can be extended to cover both WLS and GLS, and Scariano and Davenport (1984) derive an exact expression for the *Corrected F–statistic,* F^W. They show that F^W has a non–central F distribution with parameters $p,\ n-m-1$ and λ, where λ is a complex function of $\mathbf{A}$, $\hat{\boldsymbol{\beta}}_{GLS}$, $\mathbf{a}$, $\mathbf{X}$, $\mathbf{y}$ and $\mathbf{C}$, the known covariance matrix of the errors. We refer the interested reader to Scariano and Davenport (1984) for the relevant formulae.

In spite of the elegance of these results, great care must be taken with their application. The conditions under which they hold are rarely satisfied in practice, and (B.57) and (B.60) often lead to erroneous judgments concerning the significance of explanatory variables. Harvey (2017), Harvey (2019), and Boudoukh et al. (2021) illuminate the ease with which such errors of judgment arise when running batteries of tests against financial data, and describe enhancements to (B.57) and (B.60) that usefully protect against false findings of significance[12].

B.5.5 The Gauss–Markov Theorem

Under the following four assumptions:

1. The relationship between $\mathbf{y}$ and $\mathbf{X}$ is described by the linear regression model[13]: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \epsilon$,
2. The error term satisfies $\mathrm{E}\left(\epsilon \,|\, \mathbf{X}\right) = \mathbf{0}$. In econometrics textbooks, this condition is referred to as *strict exogeneity*,

[12] A variety of techniques for correcting some of the effects of multiple testing, endogeneity and overlapping observations are described and used in Chapter 12. Unfortunately, they are not exhaustive – in many settings, there is no credible way in which to correct for the collective impact of these distortions, and we are forced to use simulations to approximate the distribution of parameters under the null hypothesis. Approximate critical values for each significance level can be obtained from the simulation, and the estimated parameter can be compared to the approximate critical values to obtain an approximate significance level.

[13] Most introductory textbooks on linear regression assume that $\mathbf{X}$ is fixed and non–random, but for applications in finance and economics, the columns of $\mathbf{X}$ should be thought of as a collection of of random vectors.

3. Lack of collinearity: the rank of the data matrix is equal to $m+1$, the number of columns it contains, with probability 1, and
4. The errors are conditionally uncorrelated and their conditional variances are identical: $\mathrm{E}(\epsilon\epsilon' \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$,

the celebrated *Gauss–Markov theorem* states[14] that the OLS estimator is the *best linear unbiased estimator* (BLUE) in the sense that $\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}_{OLS}}$, the covariance matrix of its coefficient estimates, is not larger than [15] the covariance matrix of the coefficients of any other unbiased linear estimator. If we additionally assume that the errors are normally distributed, the OLS estimates are actually maximum likelihood estimates (see Section B.8.2 for details), though this is a very stringent requirement that is hardly ever satisfied in practice.

As GLS can be viewed as an application of OLS to a set of transformed variables, it inherits the virtues of OLS: it is equivalent to the maximum likelihood estimator for normally distributed errors, satisfies the Gauss–Markov theorem when Assumption 4 is changed to restrict the conditional covariance matrix of the errors only to be positive definite, and is BLUE. Additionally, $\hat{\boldsymbol{\beta}}_{GLS}$ is unbiased and consistent[16]. But, as we have earlier noted, GLS also inherits OLS's greatest liability – its lack of robustness to outliers [17].

B.6 Minimum Mean Squared Error (MMSE) Predictors

B.6.1 MMSE Predictors of a Scalar Random Variable

We earlier assumed that $\mathbf{x}$ and $\mathbf{y}$ were vectors whose entries were drawn from an unknown distribution, but we now assume that they are drawn from some known joint distribution. For simplicity, we start by considering a scalar random variable y whose value we wish to predict using a vector of predictors $\mathbf{x}$. In general, a predictor for y based on $\mathbf{x}$ can be written as

$$\hat{y} = g(\mathbf{x}), \tag{B.61}$$

where $\hat{y}$ is the predicted value of y, and $g(\cdot)$ is an arbitrary function. In many cases of practical interest in finance, y can be observed only in the future, while the components of $\mathbf{x}$ are observed at the present time, so that (B.61) is often best thought of as

$$\hat{y}_{t+1} = g(\mathbf{x}_t). \tag{B.62}$$

Once y has been observed, we can compute the prediction error

[14] See Chapter 1 of Hayashi (2000) or Chapter 4 of Greene (2018) for a proof of the Gauss–Markov theorem.

[15] An $n \times n$ matrix $\mathbf{A}_1$ is said to be smaller than another $n \times n$ matrix $\mathbf{A}_2$ if the difference of the two matrices, $\Delta\mathbf{A} = \mathbf{A}_2 - \mathbf{A}_1$, is positive definite,

[16] Hansen (2021) generalizes the Gauss–Markov theorem and shows that the OLS and GLS estimators are actually the best unbiased estimators of the regression coefficients (BUE), and that there is no need to restrict attention to the class of linear estimators. In fact, he shows that any estimator of the regression coefficients which has a smaller covariance matrix must necessarily be biased. The qualification that the estimator be unbiased is important – it has long been known that some estimators, such as the James and Stein (1961) estimator, allow a small increase in bias in return for a much larger decrease in variance. That said, unbiased estimators continue to play a central role in estimation, as they are far more easily analyzed than their biased counterparts.

[17] Efron and Morris (1973) explore Stein's estimator in an Empirical Bayes setting, and Efron and Morris (1977) provide a very readable account of the estimator and its associated paradox, and apply it to baseball scores in the 1970 season as well as to the incidence of toxoplasmosis in El Salvador.

$$e = y - \hat{y}, \tag{B.63}$$

where y, with some abuse of notation, denotes the realization of the random variable in (B.63).[18] The *mean squared prediction error* conditioned on a particular value of $\mathbf{x}$ is

$$\begin{aligned} \mathrm{MSE}_{\mathbf{x}} &= \mathrm{E}_y \left[e^2 \,|\, \mathbf{x} \right] \\ &= \mathrm{E}_y \left[(y - \hat{y})^2 \,|\, \mathbf{x} \right] \\ &= \mathrm{E}_y \left[(y - g(\mathbf{x}))^2 \,|\, \mathbf{x} \right], \end{aligned} \tag{B.64}$$

and the *Minimum Mean Squared Error Predictor*, $g_0(\mathbf{x})$, which minimizes (B.64), is given by

$$g_0(\mathbf{x}) = \mathrm{E}_y \left[y \,|\, \mathbf{x} \right], \tag{B.65}$$

where $\mathrm{E}_y \left[y \,|\, \mathbf{x} \right]$ is the conditional mean of the random variable y conditioned on $\mathbf{x}$. This can be seen by writing

$$\mathrm{E}_y \left[(y - g(\mathbf{x}))^2 \,|\, \mathbf{x} \right] = \int (y - g(\mathbf{x}))^2 f_{y|\mathbf{x}} \, dy \tag{B.66}$$

where $f_{y|\mathbf{x}}$ is the conditional density of y. If we differentiate (B.66) with respect to $g(\mathbf{x})$[19] and set the resulting expression to 0, we get

$$2 \int (g_0(\mathbf{x}) - y) f_{y|\mathbf{x}} \, dy = 0. \tag{B.67}$$

The first term in (B.67) equals $g_0(\mathbf{x})$,and it follows that

$$\begin{aligned} g_0(\mathbf{x}) &= \int y f_{y|\mathbf{x}} \, dy \\ &= \mathrm{E}_y \left[y \,|\, \mathbf{x} \right]. \end{aligned} \tag{B.68}$$

Differentiating (B.67) a second time makes clear that the second derivative is strictly positive so that (B.68) represents a minimum, and not a stationary point. Since the conditional mean predictor $g_0(\mathbf{x})$ minimizes the conditional MSE for every $\mathbf{x}$, it must also minimize the unconditional MSE. To see that this is true, observe that the unconditional MSE of the predictor is

$$\begin{aligned} \mathrm{E}_{\mathbf{x}} \left[\mathrm{MSE}_{\mathbf{x}} \right] &= \mathrm{E}_{\mathbf{x}} \left[\mathrm{E}_y \left[(y - g_0(\mathbf{x}))^2 \,|\, \mathbf{x} \right] \right] \\ &= \int \cdots \int (y - g_0(\mathbf{x}))^2 f_{y,\mathbf{x}} \, dy \, d\mathbf{x}. \end{aligned} \tag{B.69}$$

But $f_{y,\mathbf{x}}$ is non-negative, and it follows that minimizing (B.66) for each value of $\mathbf{x}$ simultaneously minimizes (B.69). While we could, in principle, have started with the objective of minimizing the unconditional MSE, the path we have chosen greatly enhances the clarity of the exposition of this important result.

The MMSE predictor is unbiased in that the expected value of its prediction is identical to the expected value of the random variable y. The notion of unbiasedness presented in subsection B.4.1, in contrast, is concerned with bias in the estimation of an unknown, but constant, parameter.

Remarkably, the prediction error of the MMSE predictor is orthogonal to *any* function of $\mathbf{x}$, i.e.

[18] It is the norm in most probability textbooks to use capital letters for random variables and small letters for their realizations. This textbook, however, has random variables, random vectors and random matrices, as well as their realizations, to represent, and consistency in one place often opens up an inconsistency in another, or necessitates the use of obscure and clumsy notation. We therefore take the liberty of using the same letter to denote both the random variable and its realization, and where needed for clarity to distinguish a random variable from its realized value, we will do so in a footnote.

[19] For any fixed value of $\mathbf{x}$, $g(\mathbf{x})$ is a scalar, and can be thought of as a parameter with respect to which the integral is differentiated.

$$\begin{aligned}\mathrm{E}_{y,\mathbf{x}}\left[\left(y-g_0\left(\mathbf{x}\right)\right)h\left(\mathbf{x}\right)\right] &= \mathrm{E}_{\mathbf{x}}\left[E_{y|\mathbf{x}}\left[\left(y-g_0\left(\mathbf{x}\right)\right)h\left(\mathbf{x}\right)|\mathbf{x}\right]\right] \\ &= \mathrm{E}_{\mathbf{x}}\left[\left(\mathrm{E}_{y|\mathbf{x}}\left[y\right]-\mathrm{E}\left[y|\mathbf{x}\right]\right)h\left(\mathbf{x}\right)|\mathbf{x}\right] \\ &= \mathrm{E}_{\mathbf{x}}\left[\left(\mathrm{E}\left[y|\mathbf{x}\right]-\mathrm{E}\left[y|\mathbf{x}\right]\right)h\left(\mathbf{x}\right)|\mathbf{x}\right] \\ &= 0. \end{aligned} \tag{B.70}$$

B.6.2 Linear MMSE Predictors of a Scalar Random Variable

Even though (B.65) is true in general, it is not always easy to implement – $f_{y|\mathbf{x}}$ is not usually known, and even if it is, it may be non–linear, time–varying, and computationally taxing. A widely used solution to these problems is to restrict $g\left(\mathbf{x}\right)$ to be an affine function of $\mathbf{x}$, and to write

$$\hat{y} = \alpha + \mathbf{x}'\boldsymbol{\beta} \tag{B.71}$$

where α is a constant, $\mathbf{x}$ is the known observable and $\boldsymbol{\beta}$ is a vector of unknown coefficients[20].

As in our earlier development of OLS, even though (B.71) is affine in $\mathbf{x}$, we can incorporate some, but not all, of the nonlinearity that (B.61) admits: the elements of $\mathbf{x}$ can be non–linear transformations, and even cross–products, of the raw data being used to forecast y. In many cases of practical interest, this simple modification significantly lowers the mean squared prediction error of the forecasts generated by (B.71).

In the special case when $\mathbf{x}$ is jointly normally (or more generally, jointly elliptically) distributed with a finite variance, $\mathrm{E}\left(y|\mathbf{x}\right)$ is given exactly by (B.71)[21], but this is not very useful for asset returns because of their known non–normality[22]. Instead, by minimizing

$$\mathrm{E}\left[\left(y-\alpha-\mathbf{x}'\boldsymbol{\beta}\right)^2\right], \tag{B.72}$$

we show in subsection B.6.3 that the linear MMSE predictor of y is obtained by setting

$$\boldsymbol{\beta}_{LMMSE} = \mathbf{C}^{-1}\boldsymbol{\sigma} \tag{B.73a}$$

$$\alpha_{LMMSE} = \mathrm{E}\left[y\right] - \mathrm{E}\left[\mathbf{x}\right]'\boldsymbol{\beta}_{LMMSE}, \tag{B.73b}$$

where $\mathbf{C}$ is the covariance matrix of $\mathbf{x}$ and $\boldsymbol{\sigma} = \mathrm{cov}\left(y,\mathbf{x}\right)$.

In the special case when $\mathbf{x}$ is a scalar, denoted by x,

$$\hat{y} = \mathrm{E}\left[y\right] + \frac{\mathrm{cov}\left(x,y\right)}{\sigma_x^2}\left(x-\mathrm{E}\left[x\right]\right). \tag{B.74}$$

The minimum mean squared error of the Linear MMSE predictor is

$$\begin{aligned}\mathrm{MMSE} &= \sigma_y^2 - \boldsymbol{\sigma}'\mathbf{C}^{-1}\boldsymbol{\sigma} \\ &= \sigma_y^2\left(1-R_{y,\mathbf{x}}^2\right), \end{aligned} \tag{B.75}$$

[20] For clarity of exposition, $\boldsymbol{\beta}$ reflects only the weights applied to the random variables – the constant α is treated separately. In our earlier development of OLS, in contrast, our problem formulation was greatly simplified by making α the first entry in $\boldsymbol{\beta}$.

[21] See Cambanis et al. (1981) and Ding (2016) for a proof.

[22] Azzalini and Capitanio (2014) is a useful reference for non–normal and skewed distributions.

where

$$R^2_{y,\mathbf{x}} = \frac{\boldsymbol{\sigma}' \mathbf{C}^{-1} \boldsymbol{\sigma}}{\sigma_y^2} \tag{B.76}$$

is the fraction of the variance of y explained by $\mathbf{x}$.

The LMMSE predictor is clearly unbiased. That said, it is important to keep in mind that α, $\boldsymbol{\beta}$ and $\mathbf{C}$ are usually not known, but must be estimated from data. While a rich body of work on optimal estimators exists for data drawn from known distributions, most classical estimators are sensitive to misspecification and to outliers, necessitating the use of robust estimators such as `lmrobdetMM` and `covRob` in the R package `RobStatTM` for many problems in finance and economics.

B.6.3 Optimality and MSE of the LMMSE Predictor

To prove the optimality of the LMMSE predictor, we follow much the same sequence of steps we used in B.6.1. The prediction error is now given by

$$e = y - \alpha - \mathbf{x}'\boldsymbol{\beta}.$$

and the *unconditional*[23] mean squared prediction error becomes

$$\begin{aligned} \text{MSE} &= \mathrm{E}_{y,\mathbf{x}}\left[(y - \alpha - \mathbf{x}'\boldsymbol{\beta})^2\right] \\ &= \int \cdots \int (y - \alpha - \mathbf{x}'\boldsymbol{\beta})^2 \, f_{y,\mathbf{x}} \, dy \, d\mathbf{x}. \end{aligned} \tag{B.77}$$

Differentiating (B.77) with respect to α, evaluating the integrals, and set the resulting expression to 0 gives

$$\alpha = \mathrm{E}[y] - \mathrm{E}[\mathbf{x}]' \, \boldsymbol{\beta}. \tag{B.78}$$

To obtain $\boldsymbol{\beta}$, we substitute (B.78) into (B.77) to get

$$\begin{aligned} \text{MSE} &= \int \cdots \int \left((y - \mathrm{E}[y]) - (\mathbf{x} - \mathrm{E}[\mathbf{x}])' \, \boldsymbol{\beta}\right)^2 f_{y,\mathbf{x}} dy d\mathbf{x}. \\ &= \int \cdots \int \left((y - \mathrm{E}[y])^2 - 2\,(y - \mathrm{E}[y])\,(\mathbf{x} - \mathrm{E}[\mathbf{x}])' \, \boldsymbol{\beta} + \boldsymbol{\beta}' \, (\mathbf{x} - \mathrm{E}[\mathbf{x}])\,(\mathbf{x} - \mathrm{E}[\mathbf{x}])' \, \boldsymbol{\beta}\right) f_{y,\mathbf{x}} dy d\mathbf{x} \\ &= \sigma_y^2 - 2\boldsymbol{\sigma}' \, \boldsymbol{\beta} + \boldsymbol{\beta}' \, \mathbf{C} \, \boldsymbol{\beta} \end{aligned} \tag{B.79}$$

where $\boldsymbol{\sigma}$ and $\mathbf{C}$ are defined immediately below (B.73a) and (B.73b).

The gradient of (B.79), which must be $\mathbf{0}$ when MSE is minimized, is

$$\nabla\left(\sigma_y^2 - 2\boldsymbol{\sigma}' \, \boldsymbol{\beta} + \boldsymbol{\beta}' \, \mathbf{C} \, \boldsymbol{\beta}\right) = -2\boldsymbol{\sigma} + 2\mathbf{C}\boldsymbol{\beta}. \tag{B.80}$$

Additionally, the Hessian of the MSE is

$$\mathbf{H}\left(\sigma_y^2 - 2\boldsymbol{\sigma}' \, \boldsymbol{\beta} + \boldsymbol{\beta}' \, \mathbf{C} \, \boldsymbol{\beta}\right) = 2\mathbf{C}. \tag{B.81}$$

[23] We focus on the unconditional MSE because we want α and $\boldsymbol{\beta}$ to be fixed constants that, unlike $\mathrm{E}(y|\mathbf{x})$, do not depend on the realization of $\mathbf{x}$.

which is positive definite, implying that the stationary point is a minimum, and that

$$\boldsymbol{\sigma} = \mathbf{C}\,\boldsymbol{\beta} \tag{B.82}$$

and

$$\boldsymbol{\beta} = \mathbf{C}^{-1}\,\boldsymbol{\sigma}. \tag{B.83}$$

It follows that

$$\begin{aligned} \mathrm{MSE} &= \sigma_y^2 - \boldsymbol{\beta}'\,\mathbf{C}\,\boldsymbol{\beta} \\ &= \sigma_y^2 - \boldsymbol{\sigma}'\,\mathbf{C}^{-1}\,\boldsymbol{\sigma}. \end{aligned} \tag{B.84}$$

The second term on the right hand side of (B.84) can be further simplified using (B.82) and (B.83)

$$\begin{aligned} \boldsymbol{\sigma}'\,\mathbf{C}^{-1}\,\boldsymbol{\sigma} &= \frac{\left(\boldsymbol{\sigma}'\,\mathbf{C}^{-1}\,\boldsymbol{\sigma}\right)^2}{\boldsymbol{\sigma}'\,\mathbf{C}^{-1}\,\boldsymbol{\sigma}} \\ &= \frac{(\boldsymbol{\sigma}'\,\boldsymbol{\beta})^2}{\boldsymbol{\beta}'\,\mathbf{C}\,\boldsymbol{\beta}} \\ &= \frac{\mathrm{cov}\,(y, \mathbf{x}'\,\boldsymbol{\beta})^2}{\mathrm{var}\,(\mathbf{x}'\,\boldsymbol{\beta})} \\ &= \sigma_y^2 \cdot \mathrm{cor}\,(y, \mathbf{x}'\,\boldsymbol{\beta})^2 \\ &= \sigma_y^2 \cdot \mathrm{cor}\,(y, \alpha + \mathbf{x}'\,\boldsymbol{\beta})^2 \\ &= \sigma_y^2 \cdot \mathrm{cor}\,(y, \hat{y})^2 \\ &= \sigma_y^2 \cdot R_{y,\mathbf{x}}^2, \end{aligned} \tag{B.85}$$

so that (B.84) simplifies to

$$\mathrm{MSE} = \sigma_y^2 \left(1 - R_{y,\mathbf{x}}^2\right). \tag{B.86}$$

B.6.4 Conditional Distribution of a Multivariate Normal Distribution

Given the central role that the conditional mean plays in MMSE prediction of scalar random variables, it is natural to ask if one can compute the conditional mean of a random vector. In many cases of interest, we can. Let

$$\mathbf{x} = \left[\frac{\mathbf{x}_1}{\mathbf{x}_2}\right] \tag{B.87}$$

be a vector with $n_1 + n_2$ elements that is partitioned into two sub–vectors: $\mathbf{x}_1$, the upper portion, has n_1 elements, and $\mathbf{x}_2$, the lower part, has n_2 elements. The distribution of $\mathbf{x}$ is assumed to have a mean vector and covariance matrix

$$\boldsymbol{\mu} = \left[\frac{\boldsymbol{\mu}_1}{\boldsymbol{\mu}_2}\right] \tag{B.88a}$$

$$\boldsymbol{\Sigma} = \left[\begin{array}{c|c} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \hline \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{array}\right]. \tag{B.88b}$$

The dimensions of the partitioned vector and matrix must be consistent with the dimensions of $\mathbf{x}_1$ and $\mathbf{x}_2$. In the special case when the distribution of $\mathbf{x}$ is multivariate normal with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, its pdf is given by

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^{n_1+n_2} det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right). \tag{B.89}$$

The unconditional distributions of both $\mathbf{x}_1$ and $\mathbf{x}_2$ are also multivariate normal, and their pdfs are

$$f(\mathbf{x}_1) = \frac{1}{\sqrt{(2\pi)^{n_1} det(\boldsymbol{\Sigma}_{11})}} \exp\left(-\frac{1}{2}(\mathbf{x}_1-\boldsymbol{\mu}_1)\boldsymbol{\Sigma}_{11}^{-1}(\mathbf{x}_1-\boldsymbol{\mu}_1)\right) \tag{B.90}$$

and

$$f(\mathbf{x}_2) = \frac{1}{\sqrt{(2\pi)^{n_2} det(\boldsymbol{\Sigma}_{22})}} \exp\left(-\frac{1}{2}(\mathbf{x}_2-\boldsymbol{\mu}_2)\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2-\boldsymbol{\mu}_2)\right), \tag{B.91}$$

while the distribution of $\mathbf{x}_2$ conditioned on $\mathbf{x}_1$ is multivariate normal with a conditional mean vector and covariance matrix

$$\boldsymbol{\mu}_{2|1} = \boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}(\mathbf{x}_1-\boldsymbol{\mu}_1) \tag{B.92a}$$

$$\boldsymbol{\Sigma}_{2|1} = \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}. \tag{B.92b}$$

The conditional distribution of $\mathbf{x}_2$ can also be derived using the fact that it is the ratio of the joint distribution of $\mathbf{x}_1$ and $\mathbf{x}_2$ (i.e., the distribution of $\mathbf{x}$) to the marginal distribution of $\mathbf{x}_1$, i.e.

$$f_{2|1}(\mathbf{x}_2) = \frac{f(\mathbf{x})}{f(\mathbf{x}_1)}, \tag{B.93}$$

and can be shown to be[24]

$$f_{2|1}(\mathbf{x}_2) = \frac{1}{\sqrt{(2\pi)^{n_2} det(\boldsymbol{\Sigma}_{2|1})}} \exp\left(-\frac{1}{2}\left(\mathbf{x}_2-\boldsymbol{\mu}_{2|1}\right)\boldsymbol{\Sigma}_{21}^{-1}\left(\mathbf{x}_2-\boldsymbol{\mu}_{2|1}\right)\right). \tag{B.94}$$

In fact, (B.92a) is remarkably general: it gives the conditional mean[25] for all elliptical distributions with finite means, such as the t distribution with 2 or more degrees of freedom. The proof of this result is complex, and we refer the interested reader to Cambanis et al. (1981) and Ding (2016) for the details.

B.7 Estimators via the Plug-In Principle

The *plug–in principle* provides a way to obtain a non–parametric estimator of a parameter or other quantity from the true but unknown value of the parameter or other quantity as described in terms of an integral with respect to a probability density or distribution function. This principle is nicely described in Chapter 4 of Efron and Tibshirani (1993), and we provide a brief description of it here.

[24] An accessible derivation of this result can be found at `https://stats.stackexchange.com/questions/30588/deriving-the-conditional-distributions-of-a-multivariate-normal-distribution` as well as in Section 3.2 of Mardia et al. (1979).

[25] Unfortunately, (B.92b) does not correspondingly give the conditional covariance matrix for all elliptical distribtions.

The true but unknown value of a parameter θ for a random variable r with probability density $f(r)$ and distribution function $F(r)$ can be written in the functional form

$$\theta(F) = \int g(r; F)\, dF(r) = \int g(r; F)\, f(r)\, dr \tag{B.95}$$

for some appropriate function $g(r)$. For example, by choosing $g(r; F) = r$, the mean of the distribution may be written as

$$\mu(F) = \int r\, dF(r) \tag{B.96}$$

and by choosing $g(r; F) = (r - \mu(F))^2$, the variance can be written as

$$\sigma^2(F) = \int (r - \mu(F))^2\, dF(r). \tag{B.97}$$

The standard deviation functional, is of course, just

$$\sigma(F) = \sqrt{\int (r - \mu(F))^2\, dF(r)}. \tag{B.98}$$

Using the functional representation of a parameter we can obtain a parameter estimator by substituting the empirical distribution function F_n for the true, but unknown, distribution function F. The empirical distribution function (e.d.f.) is a discrete probability function that assigns a probability $1/n$ to each of the observed values $r_1, r_2, \cdots, r_n$. So for the general case we have the plug–in estimator

$$\begin{aligned}
\hat{\theta} &= \theta(F_n) \\
&= \int g(r; F_n)\, dF_n(r) \\
&= \sum_{i=1}^{n} g(r_i; F_n)\, P(r = r_i) \\
&= \frac{1}{n} \sum_{i=1}^{n} g(r_i; F_n).
\end{aligned} \tag{B.99}$$

Applying the plug–in principle to the mean functional gives the sample mean

$$\hat{\mu} = \bar{r} = \frac{1}{n} \sum_{i=1}^{n} r_i, \tag{B.100}$$

and applying it to the variance functional gives the sample variance

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{t=1}^{n} (r_t - \bar{r})^2. \tag{B.101}$$

B.8 Maximum Likelihood Estimators

In this section we begin by discussing maximum–likelihood estimators (MLEs) of location and scale parameters, then extend the discussion to the case of linear regression model coefficient estimators.

B.8.1 Location and Scale MLEs

Assume we have N i.i.d. observations $\mathbf{r} = (r_1, r_2, \cdots, r_n)$ of a random variable R whose pdf is drawn from the location–scale family. The returns can be be written as

$$r_i = \mu + s \cdot \varepsilon_i, \quad i = 1, 2, \ldots, n. \tag{B.102}$$

The pdf of each of observation is given by (B.1b), and their joint pdf $f(\mathbf{r}; \mu, s)$ is the product of their densities. The *likelihood function* $L(\mu, s)$ is defined to be the joint density function of the observations, with $\mathbf{r}$ fixed at the observed data values and with s and μ variable, so that

$$L(\mu, s) = \prod_{i=1}^{N} \frac{1}{s} f\left(\frac{r_i - \mu}{s}\right). \tag{B.103}$$

It is common practice in theoretical statistics to focus instead on the log–likelihood function $l(\mu, s)$, defined using the natural logarithm, as:

$$\begin{aligned} l(\mu, s) &= \ln L(\mu, s) \\ &= \ln\left(\prod_{i=1}^{N} \frac{1}{s} f\left(\frac{r_i - \mu}{s}\right)\right) \\ &= -N \cdot \ln(s) + \sum_{i=1}^{N} \ln f\left(\frac{r_i - \mu}{s}\right). \end{aligned} \tag{B.104}$$

The maximum likelihood estimates, or MLE's, $(\hat{\mu}, \hat{s})_{mle}$ are the values of μ and s that maximize $l(\mu, s)$ and thus maximize the probability of $\mathbf{r}$ being observed:

$$(\hat{\mu}, \hat{s})_{mle} = \underset{\mu, s}{\operatorname{argmax}}\left(-N \cdot \ln(s) + \sum_{i=1}^{N} \ln f\left(\frac{r_i - \mu}{s}\right)\right). \tag{B.105}$$

Even though (B.105) usually represents a challenging non–convex numerical optimization problem, we can gain a useful understanding of the mechanics of maximum likelihood estimation by acting as if s is known, and maximizing (B.105) with respect to μ. In this case we can ignore $N \cdot \ln(s)$ in (B.126), and minimize the negative of the second term:

$$\hat{\mu}_{mle} = \underset{\mu}{\operatorname{argmin}}\left(-\sum_{i=1}^{N} \ln f\left(\frac{r_i - \mu}{s}\right)\right). \tag{B.106}$$

If we define $\rho(x) = -\ln f(x)$ to be a *loss function*, the above MLE can be written as

$$\hat{\mu}_{mle} = \underset{\mu}{\operatorname{argmin}}\left(\sum_{i=1}^{N} \rho\left(\frac{r_i - \mu}{s}\right)\right), \tag{B.107}$$

which expresses the MLE of μ as the value that minimizes a sum of loss functions.

If we differentiate the right hand side of (B.107) with respect to μ and equate the result to 0, we obtain the estimating equation

$$\sum_{i=1}^{N} \psi\left(\frac{r_i - \mu}{s}\right) = 0, \tag{B.108}$$

where $\psi(\cdot)$ is the derivative of $\rho(x)$. It is easily seen that

$$\begin{aligned}\psi(x) &= \frac{d}{dx}\rho(x) \\ &= -\frac{d}{dx}\ln f(x) \\ &= -\frac{f'(x)}{f(x)}.\end{aligned} \tag{B.109}$$

The above form of $\psi(x)$[26] is known as the *score function* in mathematical statistics, and measures the sensitivity of the likelilihood to small changes in the estimated parameter. It attains a value of 0 at a local minimum of the loss function, allowing the identification of the parameter value that maximizes the likelihood function.

We now discuss two important distributions that admit simple solutions to (B.108), and a third one which does not, but is none the less an important distribution for modeling fat–tailed asset returns.

Example B.1. Normal Distribution MLE.

In the case of the standard normal pdf we have:

$$f(x) = \frac{1}{\sqrt{2\pi}} exp\left(-x^2/2\right) \tag{B.110}$$

and

$$\rho_{mle,normal}(x) = \ln(2\pi)/2 + x^2/2 \tag{B.111}$$

so that

$$\psi_{mle,normal}(x) = x. \tag{B.112}$$

In this case (B.108) reduces to

$$\sum_{i=1}^{N} r_i = N \cdot \hat{\mu}_{mle,normal}, \tag{B.113}$$

and the location estimator $\hat{\mu}_{mle,normal}$ is seen to be the sample mean. Likewise, the joint estimate $\hat{s}_{mle,normal}$ can be shown to be the sample standard deviation obtained as the square root of the biased sample variance (B.16).

Example B.2. Laplacian Distribution MLE.

The standard form of the Laplacian (symmetric two–sided exponential) distribution is:

$$f(x) = \frac{1}{2} \cdot exp\left(-|x|\right), \tag{B.114}$$

and

$$\rho_{mle,Laplacian}(x) = constant + |x|, \tag{B.115}$$

so that

$$\psi_{mle,Laplacian}(x) = sgn(x) = \begin{cases} +1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}. \tag{B.116}$$

[26] More generally, when the distribution is characterized by more than one parameter, it is the *gradient* of $\rho(x)$.

Once again, the location MLE does not depend upon s, and (B.116) makes clear that the optimal estimate of location must have the same number of observations to its left as it does to its right, so that $\hat{\mu}_{mle,Laplacian} = MED(\mathbf{r})$. It is easily shown that the median is unique for an odd sample size, and is not unique for an even sample size. By convention, it is taken to be the mean of the two middle observations when the sample size is even. It can also be shown that the scale MLE is the mean–absolute deviation about the median estimator $\hat{s}_{mle,Laplacian} = n^{-1} \cdot \sum_{i=1}^{n} |r_i - MED(\mathbf{r})|$.

Example B.3. t–Distribution MLE

In this case we assume that f is the standard t–distribution pdf with a known degrees of freedom parameter ν, and have:

$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{2\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + x^2/\nu\right)^{-(\nu+1)/2}. \tag{B.117}$$

$$\rho_{mle,tdist}(x) = constant + \frac{\nu+1}{2} log\left(1 + x^2/\nu\right) \tag{B.118}$$

$$\psi_{mle,tdist}(x) = \frac{\nu+1}{\nu} \frac{x}{1 + x^2/\nu}. \tag{B.119}$$

This is a non–convex optimization problem that requires a good numerical optimization algorithm, as well as an initial estimate of location and knowledge of ν, both of which may have to be estimated from data. Consequently, even though the t–distribution has often been proposed for modeling fat–tailed distributions that are not well modeled by a Laplacian distribution, its MLE has not seen wide usage, though good numerical algorithms to solve the MLE are found in packages such as `sn` and `MASS, both of which` are described in Section 3.5

Figure B.2 displays the loss function ρ and score functions ψ for these three examples. In the case of the t–distributions, the scale parameter s is set so that the standard deviation is $\sigma = 1$, using the known relationship $\sigma^2 = s^2 \cdot \nu/(\nu - 2)$.

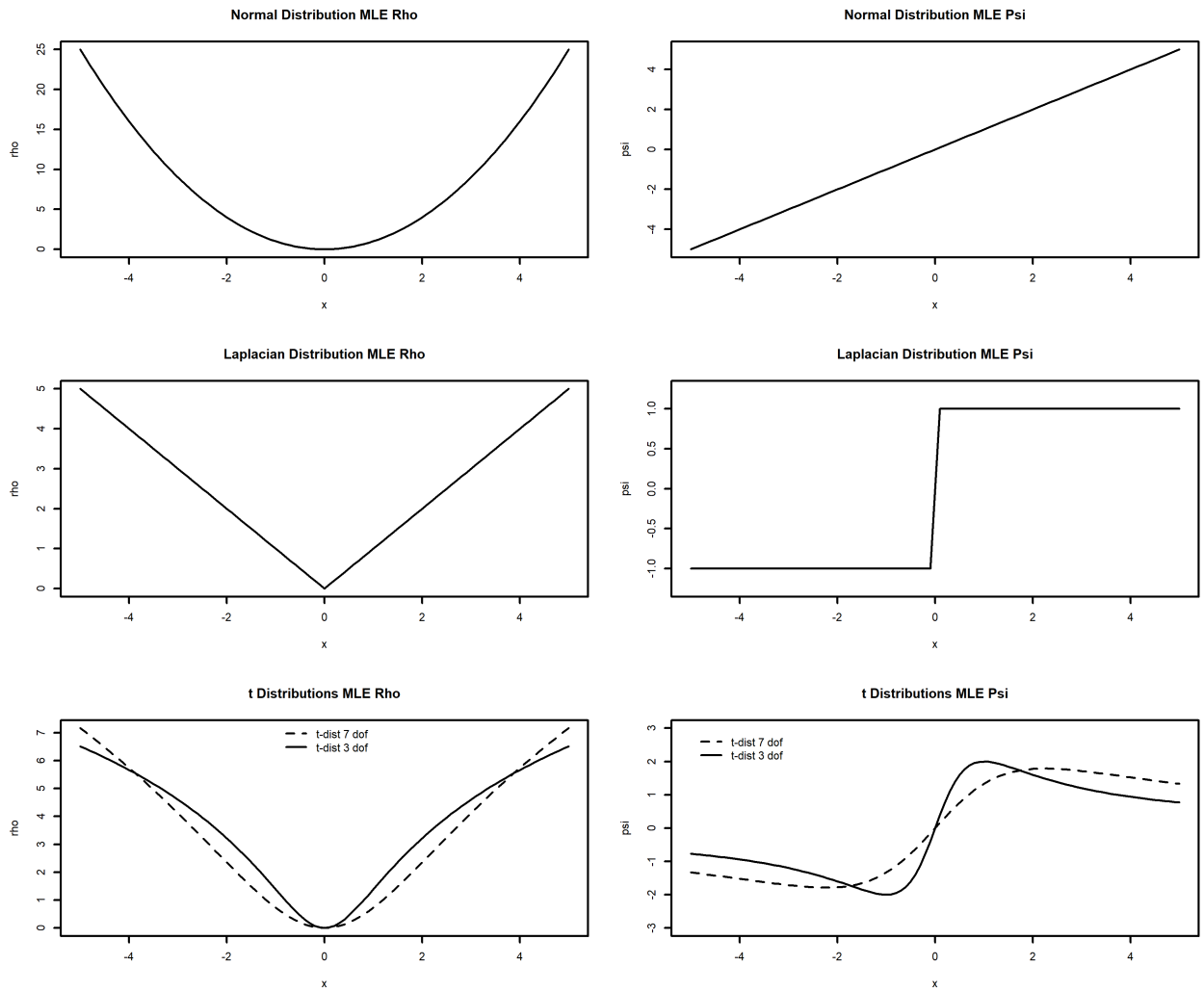

Fig. B.2 MLE Loss Functions (ρ) and Score Functions (ψ) for Normal, Laplacian, and t-Distributions

B.8.2 Regression Model MLEs

We next extend our results on MLEs to cross–section regression models. We focus on the cross–section model for a single time period with the time subscript t dropped, but keep in mind that the model will be fit for each time period t in some time window:

$$r_i = f_o + \beta_{i1} \cdot f_1 + \beta_{i2} \cdot f_2 + \cdots + \beta_{iK} \cdot f_K + \epsilon_i,\ i = 1, 2, \cdots, N. \tag{B.120}$$

Let $\mathbf{B}$ be the $N \times (K+1)$ matrix whose i^{th} row is $\mathbf{b}_i' = (1, \beta_{i1}, \beta_{i2}, \cdots, \beta_{iK})$ and let $\mathbf{f} = (f_o, f_1, \cdots, f_K)'$, Then the above model may be written more compactly as:

$$r_i = \mathbf{b}_i'\mathbf{f} + \epsilon_i,\ i = 1, 2, \cdots, N. \tag{B.121}$$

In the above model the β_{ik} are fixed predictor variables and the f_k are unknown coefficients, and for this single time–period model the f_k can be considered as unknown regression coefficient parameters,

with f_o being the intercept[27]. We make the standard assumption that the errors ϵ_i are independent and identically distributed (i.i.d.) random variables that are also independent of the β_{ik}, and further assume that they have a pdf of the scale family form

$$p(\varepsilon) = \frac{1}{s} p_0\left(\frac{\varepsilon}{s}\right),\ s > 0 \tag{B.122}$$

where p_0 is a standard symmetric density such as a standard normal or standard Laplacian distribution. Using (B.121), we have

$$p(\varepsilon_i) = \frac{1}{s} p_0\left(\frac{r_i - \mathbf{b}_i'\mathbf{f}}{s}\right), \tag{B.123}$$

and since the ε_i are i.i.d., the joint probability density $f(\mathbf{r}; \mathbf{f}, s)$ of the observations $\mathbf{r} = (r_1, r_2, \cdots, r_n)$ is

$$f(\mathbf{r}; \mathbf{f}, s) = \prod_{i=1}^{N} \frac{1}{s} p_0\left(\frac{r_i - \mathbf{b}_i'\mathbf{f}}{s}\right). \tag{B.124}$$

It follows that the log–likelihood function, $l(\mathbf{f}, s)$, is:

$$\begin{aligned} l(\mathbf{f}, s) &= \ln L(\mathbf{f}, s) \\ &= -N \cdot \ln(s) + \sum_{i=1}^{N} \ln p_0\left(\frac{r_i - \mathbf{b}_i'\mathbf{f}}{s}\right), \end{aligned} \tag{B.125}$$

which is virtually identical to (B.105).

The maximum likelihood estimates $(\hat{\mathbf{f}}, \hat{s})_{mle}$ are the values of $(\mathbf{f}, s)$ that maximize the above log–likelihood, and we write this as:

$$(\hat{\mathbf{f}}, \hat{s})_{mle} = \underset{\mathbf{f}, s}{\operatorname{argmax}} \left(-N \cdot \ln(s) + \sum_{i=1}^{N} \ln p_0\left(\frac{r_i - \mathbf{b}_i'\mathbf{f}}{s}\right)\right). \tag{B.126}$$

Once again, we explore the impact of non–normal $p_0(\varepsilon)$ on the linear regression MLE by acting as if s is known, and consider the optimization only with respect to $\mathbf{f}$. By analogy to (B.106) and (B.107) we can write:

$$\hat{\mathbf{f}}_{mle} = \underset{\mathbf{f}}{\operatorname{argmin}} \left(-\sum_{i=1}^{N} \ln p_0\left(\frac{r_i - \mathbf{b}_i'\mathbf{f}}{s}\right)\right), \tag{B.127}$$

and

$$\hat{\mathbf{f}}_{mle} = \underset{\mathbf{f}}{\operatorname{argmin}} \left(\sum_{i=1}^{N} \rho\left(\frac{r_i - \mathbf{b}_i'\mathbf{f}}{s}\right)\right), \tag{B.128}$$

where $\rho(x) = -\ln\, p_0(x)$.

By taking the derivative of the right hand side of (B.128) with respect to $\mathbf{f}$, and setting it equal to zero, we can characterize $\hat{\mathbf{f}}_{mle}$ as the solution of the estimating equation

$$\sum_{i=1}^{N} \mathbf{b}_i \psi\left(\frac{r_i - \mathbf{b}_i'\mathbf{f}}{s}\right) = 0 \tag{B.129}$$

[27] In practice, the cross–section factor models are fit for each time period t, and the resulting f_{kt} are treated as random variables that define random time series as time t evolves. This aspect is discussed further in Chapter 7.

where $\psi(x) = \frac{d}{dx}\rho(x)$.

We now discuss three possible error distribution choices in terms of the pdf $p_0(\varepsilon)$, and their corresponding regression estimators.

Example B.4. Normal Distribution Regression MLE

In the case, the density p_0 is the standard normal pdf, and as (B.128) does not depend upon s, the result is the least–squares (LS) estimate

$$\hat{\mathbf{f}}_{mle,normal} = \hat{\mathbf{f}}_{LS} = \underset{\mathbf{f}}{\operatorname{argmin}} \left(\sum_{i=1}^{N} (r_i - \mathbf{b}_i' \mathbf{f})^2 \right), \tag{B.130}$$

and (B.129) becomes the corresponding LS normal equations:

$$\sum_{i=1}^{N} \mathbf{b}_i \left(r_i - \mathbf{b}_i' \hat{\mathbf{f}}_{LS} \right) = \mathbf{0}. \tag{B.131}$$

Laplacian Distribution Regression MLE

In this case, p_0 is the Laplacian pdf, so that the resulting regression estimator is the Least–Absolute–Deviation (LAD) estimator:

$$\hat{\mathbf{f}}_{mle,Laplacian} = \hat{\mathbf{f}}_{LAD} = \underset{\mathbf{f}}{\operatorname{argmin}} \left(\sum_{i=1}^{N} |r_i - \mathbf{b}_i' \mathbf{f}| \right). \tag{B.132}$$

The *LAD* estimator is also known as the *L1* estimator, and good numerical methods for solving the *LAD/L1* minimization problem (B.132) are widely available in both commercial and open source software packages.

t–Distribution Regression MLE

In this case we assume that p_0 is the standard t–distribution pdf with known parameter ν, and the t–distribution regression MLE minimization problem becomes

$$\hat{\mathbf{f}}_{mle,tdist} = \underset{\mathbf{f}}{\operatorname{argmin}} \left(\sum_{i=1}^{N} \log \left(1 + \frac{1}{\nu} \left(\frac{r_i - \mathbf{b}_i' \mathbf{f}}{s} \right)^2 \right) \right). \tag{B.133}$$

As in the case of Example (B.3), the above is a non–convex optimization problem that is not easily solved in closed form, though some of the powerful numerical solution techniques described in Chapter 18 and Azzalini and Capitanio (2014) have applicability to this problem as well.

Error Distribution Tail Fatness and Shapes of MLE Rho and Psi Functions

It is important to note the relationship between the fatness of the tail of the error distributions, and the shapes of the $\rho(x)$ and $\psi(x)$ functions. For the normal distributions, $\rho_{mle,normal}(x)$ is quadratic and grows very rapidly, while $\psi_{mle,normal}(x) = x$ is increasing and unbounded. For Laplacian distributions, which are more fat–tailed than normal distributions, $\rho_{mle,exp}(x) = |x|$ grows only linearly but is unbounded, and $\psi_{mle,exp}(x) = sgn(x)$ is non–decreasing and bounded. For t–distributions, which are more fat–tailed than the Laplacian distribution, $\rho_{mle,tdist}(x)$ grows only logarithmically (slower than linear) but is still unbounded, and $\psi_{mle,tdist}(x)$ is not only bounded but is also non–monotonic and decreases to zero as x tends to plus and minus infinity.

Joint MLE of Regression and Scale Parameters

Computing the joint MLEs $\hat{\mathbf{f}}_{mle}$ and $\hat{s}_{mle}$ of the regression and scale parameters is in general a complex nonlinear optimization for non–normal error distributions such as the t–distribution. However, in the case of the normal and Laplacian distributions, it is straightforward to derive formulas for the joint MLEs. For the normal distribution, $\hat{s}_{mle,normal}$ is the square root of the sample average of the squared residuals $\hat{\varepsilon}_i = r_i - \mathbf{b}_i'\hat{\mathbf{f}}_{mle,normal}$. For Laplacian distributions, $\hat{s}_{mle,exp}$ is the sample average of the absolute value of the residuals $\hat{\varepsilon}_i = r_i - \mathbf{b}_i'\hat{\mathbf{f}}_{mle,normal}$.

The Maximum Likelihood Principle and Asymptotic Optimality of MLEs

This *maximum–likelihood principle* asserts that if one knows the data distribution then one should use a maximum–likelihood estimator. Although this is only a reasonableness principle that is commonly invoked in the case of finite–sample estimators, it has the benefit of the following asymptotic optimality property: if the data is generated by an assumed distribution, then the MLE based on that distribution has an asymptotic variance that is the smallest asymptotic variance achievable by a consistent estimator. For a given distribution there is an analytic expression for that minimum variance, called the information–inequality lower bound. The formula for the lower bound is given in TS B.14.

Maximum Likelihood Estimators generalize naturally to outlier–resistant robust estimators, and the similarities between the problem formulations in Section B.8 and Chapter 18 are striking. In fact, a regression M-estimator is best viewed as an outlier-resistant generalization of the regression MLE equations (B.128) and (B.129).

B.8.3 Covariance Matrix MLEs

The sample covariance matrix estimator is known to be a maximum likelihood estimator when the return vectors $\mathbf{r}_t = (r_{t1}, \ldots, r_{tN})'$, $t = 1, , 2, \cdots, T$ are multivariate normal. The proof, however, is fairly complex and is typically covered only in multivariate statistics courses. Maximum–likelihood estimators for the case of non–normal multivariate elliptical distributions are discussed in Chapter 7.

B.9 Useful Inequalities

Markov's Inequality

If X is any non–negative random variable ($X \geqslant 0$), and a is a positive constant, then:

$$P(X \geqslant a) \leqslant \frac{\mathrm{E}[X]}{a}, a > 0. \tag{B.134}$$

Moment Bound

Markov's Inequality is easily extended to higher moments of X. If X is any non–negative random variable, and if a and r are positive constants, then:.

$$P(X \geqslant a) \leqslant \frac{\mathrm{E}\left[X^r\right]}{a^r},\ a, r > 0. \tag{B.135}$$

In particular,

$$P(X \geqslant a) \leqslant \inf_{r>0} \frac{\mathrm{E}\left[X^r\right]}{a^r},\ a > 0. \tag{B.136}$$

Equation B.136 is known as the *Moment Bound*, and Philips and Nelson (1995) show that the moment bound is tighter than Chernoff's Bound for positive tail probabilities.

Chebyshev's Inequality

If a random variable X has mean $\mu = \mathrm{E}[X]$ and variance $\sigma^2 = \mathrm{E}\left[(X-\mu)^2\right]$, and b is any positive constant, then:

$$P\left(|X-\mu| \geqslant b\right) \leqslant \frac{\sigma^2}{b^2},\ b > 0. \tag{B.137}$$

Chernoff's Bound

Chernoff's Bound is the tightest general purpose inequality that applies to random variables whose domain is the entire real line.

$$P(X \geqslant a) \leqslant \inf_{\theta>0} \frac{\mathrm{E}(\exp(\theta X))}{\exp(\theta a)} \tag{B.138a}$$

$$P(X \leqslant a) \leqslant \inf_{\theta<0} \frac{\mathrm{E}(\exp(\theta X))}{\exp(\theta a)}. \tag{B.138b}$$

Jensen's Inequality

The probabilistic version of Jensen's inequality for the case of a convex function g is a generalization of the statement that the secant line of a function lies below the function. Specifically, if X is a random variable with expected value $\mathrm{E}[X]$, and g is a convex function, then

$$g\left(\mathrm{E}[X]\right) \leqslant \mathrm{E}\left[g(X)\right]. \tag{B.139}$$

If g is a strictly convex function, then Jensen's inequality is strict:

$$g\left(\mathrm{E}[X]\right) < \mathrm{E}\left[g(X)\right]. \tag{B.140}$$

For a concave and strictly concave functions g, the above inequalities are reversed.

B.10 Limit Theorems for Random Variables and Distributions

This section discusses key limit theorems of sequences of random variables and distribution functions that play a central role in the large sample asymptotic theory of parameter estimators. For further background on such asymptotics, the document "Notes for a graduate level course in asymptotics for statisticians" by Professor David R. Hunter (available at `https://sites.math.rutgers.edu/~sg1108/asymp1.pdf`) is highly recommended.

B.10.1 Random Variables Convergence

Convergence in Probability

An infinite sequence of random variables $X_1, X_{2,} \cdots$ *converges in probability* to a random variable X if for every $\varepsilon > 0$ the following limit condition holds

$$\lim_{n\to\infty} P\left(|X_n - X| < \varepsilon\right) = 1, \tag{B.141}$$

and in that case we write

$$X_n \xrightarrow{p} X. \tag{B.142}$$

Sometimes X will be a constant $X = c$, and frequently the random variables X_n are parameter estimates $X_n = \hat{\theta}_n$.

Weak Law of Large Numbers

If $X_1, X_{2,} \cdots$ are independent and identically distributed with finite mean $\mu = \mathrm{E}(X_1)$, then the sample mean $\overline{X}_n = \dfrac{1}{n}\sum_{i=1}^{n} X_i$ converges in probability to μ:

$$\overline{X}_n \xrightarrow{p} \mu. \tag{B.143}$$

The proof of the result as stated is involved. But a proof under the assumption that the X_i have finite variance σ^2, which is often a reasonable assumption, is an easy consequence of Chebychev's inequality. Additionally, it is not hard to show that for i.i.d. X_i with finite variance σ^2, the quantity $n^{-1}\sum_{i=1}^{n}(X_i - \mu)^2$ converges in probability to σ^2.

B.10.2 Convergence in Distribution

Let the infinite sequence of random variables $X_1, X_2, \cdots$ have cumulative distribution functions $F_1(x), F_2(x), \cdots$,and let X have cumulative distribution function $F(x)$, and let C be the set of points of continuity of $F(x)$. The sequence $X_1, X_{2,} \cdots$ is said to *converge in distribution* X if

$$\lim_{n\to\infty} F_n(x) = F(x),\ x \in C \tag{B.144}$$

and one writes this as:

$$X_n \xrightarrow{d} X \quad \text{or} \quad X_n \xrightarrow{d} F. \tag{B.145}$$

NOTE: For our purposes in this book the reader will only need the above two types of convergence of random variables. However, there are other modes of convergence of random variables, e.g., convergence almost surely and convergence in mean, and the interested reader can find a good discussion of these in the Hunter reference given at the beginning of this section.

B.10.3 Slutsky's Theorem

If $X_1, X_2 \cdots$ and $Y_1, Y_2 \cdots$ are infinite sequences of random variables such that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, where c is a constant, then:

$$X_n + Y_n \xrightarrow{d} X + c, \qquad X_n Y_n \xrightarrow{d} cX \qquad X_n / Y_n \xrightarrow{d} X/c, \text{ provided } c \neq 0.$$

B.10.4 Central Limit Theorem (CLT)

Let $X_1, X_2, \cdots$ be an infinite sequence of independent and identically distributed (i.i.d.) random variables with finite mean μ, finite variance σ^2, and let $\overline{X}_n$ be the sample mean of the X_i for a sample of size n. There exist a number of sophisticated *central limit theorems* (CLTs) asserting the asymptotic normality of the sample mean, the simplest of which is the following classical CLT:

$$\sqrt{n}\left(\overline{X}_n - \mu\right) \xrightarrow{d} N\left(0, \sigma^2\right) \quad \text{or} \quad \sqrt{n}\left(\overline{X}_n - \mu\right) \xrightarrow{d} Y \tag{B.146}$$

where Y is normally distributed with mean zero and variance σ^2. Equivalently, this CLT can be stated as:

$$\sqrt{n}\frac{\overline{X} - \mu}{\sigma} \xrightarrow{d} N(0, 1). \tag{B.147}$$

B.11 Estimator Asymptotic Results and Their Use

B.11.1 Consistent Estimators

A *consistent* estimator $\hat{\theta}_n$ is one that *converges in probability* to the true but unknown parameter value θ as the sample size n goes to infinity, and we write this property of an estimator as:

$$\hat{\theta}_n \xrightarrow{p} \theta. \tag{B.148}$$

We provide a precise mathematical definition of convergence in probability in Section B.10.1, but a verbal description of this property is as follows: as the sample size tends to infinity, the probability that $\left|\hat{\theta}_n - \theta\right|$ is smaller than any specified positive amount converges to 1.

Most estimators commonly used in practice are consistent under standard assumptions. For example, the sample mean is a consistent estimator of the mean μ, the sample variance estimators $\hat{\sigma}_n^2$ and S_n^2 are both consistent estimators of the variance σ^2, and the sample standard deviation estimator $\hat{\sigma}_n$ is a consistent estimator of the standard deviation σ.

B.11.2 Estimator Asymptotic Variance

Consider the sample mean

$$\overline{r}_n = \frac{1}{n}\sum_{t=1}^{n} r_t$$

based on on uncorrelated returns r_t, $t = 1, 2, \ldots, n$ with a common mean μ and common variance σ^2. When the context is clear we will drop the subscript n on $\overline{r}_n$. It is well known that the finite sample variance of $\overline{r}_n$ is

$$V_n\left(\bar{r}\right) = \text{var}(\bar{r}) = \frac{\sigma^2}{n}.$$

and so the variance of the sample mean goes to zero at rate $1/n$.

Most well–known estimators have a variance that converges to zero at a rate $1/n$ under standard assumptions. For example, the sample median, sample covariance, sample correlation coefficient, and regression coefficient estimators, among many others, have variances that decrease at a rate $1/n$. For any such estimator $\hat{\theta}_n$, one can obtain an "asymptotic variance" by applying a multiplicative factor $\sqrt{n}$ to the estimator in an appropriate manner. For example, in the case of the sample mean $\bar{r}_n$ one has $\text{var}\left(\sqrt{n}\,\bar{r}_n\right) = \sigma^2$ for every sample size n.

If an estimator $\hat{\theta}_n$ is a consistent estimator of a parameter θ, then under certain standard conditions satisfied by most estimators encountered in practice, the distribution of the standardized estimator $\sqrt{n}\left(\hat{\theta}_n - \theta\right)$ converges in distribution to a normal distribution that has mean zero and an asymptotic variance $V\left(\theta; \hat{\theta}_n\right)$, which is written as

$$\sqrt{n}\left(\hat{\theta}_n - \theta\right) \xrightarrow{d} N\left(0, V\left(\theta; \hat{\theta}_n\right)\right). \tag{B.149}$$

The argument $\hat{\theta}_n$ in $V\left(\theta; \hat{\theta}_n\right)$ reflects that fact that the asymptotic variance depends on the choice of estimator, and when the context makes it clear which estimator is being considered, the asymptotic variance is written more simply as $V(\theta)$.

Example B.5. Asymptotic Variance of the Sample Variance Estimator

If i.i.d. returns r_i have a finite fourth moment, then it turns out that the asymptotic variance of the sample variance estimator $\hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^n (r_i - \bar{r}_n)^2$ is given by

$$V(\hat{\sigma}_n^2) = \text{var}\left[(r_1 - \mu)^2\right] = \mu_4 - \sigma^4 \tag{B.150}$$

where $\mu_4 = \text{E}\left[(r - \mu)^4\right]$ is the fourth central moment. A proof of the this result is as follows.

Let $a = \text{var}\,(r_1 - \mu)^2$, and note that:

$$\begin{aligned} a &= \text{E}\left[(r_1 - \mu)^4\right] - \text{E}^2\left[(r_1 - \mu)^2\right] \\ &= \mu_4 - \sigma^4. \end{aligned}$$

Since subtracting μ from both r_i and $\bar{r}$ does not change the distribution of $r_i - \bar{r}$, we can assume the mean of the r_i is zero. Consequently

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^n (r_i - \bar{r}_n)^2 &= \frac{1}{n}\sum_{i=1}^n r_i^2 - \frac{2}{n}\sum_{i=1}^n r_i\bar{r}_n + \bar{r}_n^2 \\ &= \frac{1}{n}\sum_{i=1}^n r_i^2 - \bar{r}_n^2. \end{aligned}$$

and recalling that the mean of the r_i is zero, we have.

$$\sqrt{n}\left(\hat{\sigma}_n^2 - \sigma^2\right) = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n r_i^2 - \sigma^2\right) - \sqrt{n}\,\bar{r}_n^2.$$

It follows from the CLT that the first term on the right converges in distribution to a $N(0, a)$ distribution. Furthermore, the second term can be written as $\bar{r}_n \sqrt{n}\,\bar{r}_n$, and, by the CLT, $\sqrt{n}\,\bar{r}_n \xrightarrow{d} Z$ where Z is a $N(0,1)$ random variable, and $\bar{r}_n$ converges in probability to $c = 0$. By Slutsky's Theorem the right hand side of the above equation converges to a $N(0, a)$ distribution.

B.12 Delta Methods

We first discuss the univariate delta method that allows one to obtain the asymptotic variance of a nonlinear transformation of an estimator, a leading example of which is obtaining the asymptotic variance of a sample volatility estimator from the asymptotic variance of a sample variance estimator. We then discuss the bivariate delta method that that allows one to obtain the asymptotic variance of a performance estimator that is the ratio of two estimators based on the same set of returns, e.g., the Sharpe ratio and Sortino ratio, from the asymptotic 2×2 covariance matrix of the individual estimators for the numerator and denominator.

B.12.1 Univariate Delta Method

Suppose we have a formula $V(\theta) = V(\theta; \hat{\theta}_n)$ for the asymptotic variance of a consistent estimator $\hat{\theta}_n$, and want to find the variance $V\left(\theta; g\left(\hat{\theta}_n\right)\right)$ of a nonlinear transformation $g\left(\hat{\theta}_n\right)$ of $\hat{\theta}_n$, where the function $g(t)$ has a continuous first derivative $g'(t)$. Using the Mean Value Theorem gives

$$g\left(\hat{\theta}_n\right) = g(\theta) + g'(\theta^*) \cdot \left(\hat{\theta}_n - \theta\right), \text{ where } \theta^* \text{ lies between } \theta \text{ and } \hat{\theta}_n. \tag{B.151}$$

It follows that

$$\sqrt{n}\left(g\left(\hat{\theta}_n\right) - g(\theta)\right) = g'(\theta^*) \cdot \sqrt{n}\left(\hat{\theta}_n - \theta\right), \text{ with } \theta^* \text{ between } \theta \text{ and } \hat{\theta}_n. \tag{B.152}$$

Taking the limit on the right hand side as $n \to \infty$, the consistency of $\hat{\theta}_n$ for θ implies that $\theta^* \xrightarrow{p} \theta$ and $g(\theta^*) \xrightarrow{p} g(\theta)$ as $n \to \infty$, and so the right hand side converges to $g'(\theta) \cdot Y$, where Y has mean zero and variance $V(\theta)$. Thus by Slutsky's Theorem, the right hand side converges to a random variable with mean zero and variance $[g'(\theta)]^2 \cdot V(\theta)$. It follows that the asymptotic variance of the left hand side above is:

$$V\left(\theta; g\left(\hat{\theta}_n\right)\right) = [g'(\theta)]^2 V(\theta). \tag{B.153}$$

Example B.6. Volatility Estimator Asymptotic Variance

Since volatility is related to variance through the square root transformation $\sigma = \sqrt{\sigma^2}$, the transformation $g(t) = t^{1/2}$ and $g'(t) = t^{-1/2}/2$. With $t = \theta = \sigma^2$ we have $[g'(\theta)]^2 = 1/4\sigma^2$ and use of the formula (B.150) for the asymptotic variance of the sample variance gives:

$$V(\sigma; \hat{\sigma}_n) = \frac{\mu_4 - \sigma^4}{4\sigma^2}. \tag{B.154}$$

B.12.2 The Bivariate Delta Method

Performance estimators such as the Sharpe ratio, the Sortino ratio, and the Information ratio, are ratios of two estimators, and for the purpose of the discussion here, it is assumed that both estimators use the same set of i.i.d. returns $r_1, r_2, \cdots, r_n$. For example, the Sharpe ratio estimator is

$$\widehat{\mathrm{SR}} = \frac{\hat{\mu}_e}{\hat{\sigma}} = \frac{\hat{\mu} - r_f}{\hat{\sigma}} \tag{B.155}$$

where $\hat{\mu}$ is the sample mean, and $\hat{\sigma}$ is the sample variance, of the set of returns. For purposes of the discussion here, it is assumed that the r_i are independent and identically distributed.

A performance ratio estimator $\widehat{\mathrm{PR}}$ in general has the form

$$\widehat{\mathrm{PR}} = \frac{\hat{\theta}_1}{\hat{\theta}_2}$$

where $\hat{\theta}_1$ and $\hat{\theta}_2$ are two estimators based on the same set of returns. The asymptotic variance of such an estimator can be computed using the following bivariate delta method, which is an extension of the univariate delta method in the previous sub–section.

Let $\boldsymbol{\theta} = (\theta_1, \theta_2)'$ and let $g(\boldsymbol{\theta}) = g(\theta_1, \theta_2)$ be a scalar–valued transformation of $\boldsymbol{\theta}$. Then given a bivariate estimator $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \hat{\theta}_2)'$, one obtains the estimator $\widehat{g(\boldsymbol{\theta})} = g(\hat{\theta}_1, \hat{\theta}_2)$ of $g(\boldsymbol{\theta})$. For example, in the case of the Sharpe ratio:

$$g(\boldsymbol{\theta}) = \frac{\theta_1}{\sqrt{\theta_2}} \quad \text{with} \quad \theta_1 = \mu_e, \theta_2 = \sigma^2. \tag{B.156}$$

A Taylor series expansion of $\widehat{g(\boldsymbol{\theta})}$ about the true value $\boldsymbol{\theta}$ gives:

$$\widehat{g(\boldsymbol{\theta})} = g(\boldsymbol{\theta}) + \frac{\partial g(\boldsymbol{\theta})}{\partial \theta_1}\left(\hat{\theta}_1 - \theta_1\right) + \frac{\partial g(\boldsymbol{\theta})}{\partial \theta_2}\left(\hat{\theta}_2 - \theta_2\right) + Rem_n$$

where the estimates are based on i.i.d. returns $r_1, r_2, \cdots, r_n$, and the remainder Rem_n can be shown to go to zero faster than $1/\sqrt{n}$ under standard conditions. The asymptotic variance $V_{\hat{g}} = V_{\hat{g}}(\boldsymbol{\theta})$ of $\widehat{g(\boldsymbol{\theta})}$ is the limiting asymptotic variance of

$$\sqrt{n} \cdot \left(\widehat{g(\boldsymbol{\theta})} - g(\boldsymbol{\theta})\right) = \frac{\partial g(\boldsymbol{\theta})}{\partial \theta_1}\sqrt{n}\left(\hat{\theta}_1 - \theta_1\right) + \frac{\partial g(\boldsymbol{\theta})}{\partial \theta_2}\sqrt{n}\left(\hat{\theta}_2 - \theta_2\right) + \sqrt{n} \cdot Rem_n.$$

Assuming the usual asymptotic joint normality of the estimates $\hat{\theta}_1$ and $\hat{\theta}_2$, and noting that the remainder term goes to zero asymptotically, the asymptotic variance of $\widehat{g(\boldsymbol{\theta})}$ is:

$$V_{\hat{g}} = \nabla g(\boldsymbol{\theta})' \mathbf{V}_{\boldsymbol{\theta}} \nabla g(\boldsymbol{\theta}) \tag{B.157}$$

where $\mathbf{V}_{\boldsymbol{\theta}}$ is the asymptotic covariance matrix of the estimator $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \hat{\theta}_2)'$:

$$\mathbf{V}_{\boldsymbol{\theta}} = \begin{pmatrix} \mathrm{var}_\infty(\hat{\theta}_1) & \mathrm{cov}_\infty(\hat{\theta}_1, \hat{\theta}_2) \\ \mathrm{cov}_\infty(\hat{\theta}_1, \hat{\theta}_2) & \mathrm{var}_\infty(\hat{\theta}_2) \end{pmatrix}. \tag{B.158}$$

For estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ that are sums of non–linear functions of i.i.d. returns, the above asymptotic variances and covariances can be computed from a single term of each of the two summations. This is the case in the following example.

Example B.7. The Sharpe Ratio Asymptotic Variance Using the Bivariate Delta Method.

For the Sharpe ratio we have $\hat{\theta}_1 = \bar{r} - r_f = \frac{1}{n}\sum_{t=1}^{n} r_i - r_f$ and $\hat{\theta}_2 = \frac{1}{n}\sum_{t=1}^{n}(r_i - \bar{r})^2$. The asymptotic variance of $\hat{\theta}_1$ is σ^2, and from (B.150) we know that the asymptotic variance of $\hat{\theta}_2$ is $\mu_4 - \sigma^4$. The asymptotic covariance between $\hat{\theta}_1$ and $\hat{\theta}_2$ is easily shown to be $\mu_3 = \mathrm{E}\left[(r_1 - \mu)^3\right]$. Thus

$$\mathbf{V}_\theta = \begin{pmatrix} \sigma^2 & \mu_3 \\ \mu_3 & \mu_4 - \sigma^4 \end{pmatrix}$$

and since for the Sharpe ratio $g(\theta)$ is given by (B.156), its gradient is:

$$\nabla g(\theta) = \left(\frac{1}{\sigma}, -\frac{\mu_e}{2\sigma^3}\right)'.$$

Plugging the above two expressions into (B.157) shows that the asymptotic variance of the Sharpe ratio is:

$$V_{\widehat{SR}} = 1 - K_3 \cdot SR + \frac{K_4 + 2}{4} SR^2 \tag{B.159}$$

where $K_3 = \mu_3/\sigma^3$ is the coefficient of skewness, and $K_4 = \mu_4/\sigma^4 - 3$ is excess kurtosis. [28] In the special case of normally distributed returns, where $K_3 = K_4 = 0$, the above expression reduces to:

$$\mathrm{var}_\infty(\mathrm{SR}_n) = 1 + \frac{1}{2}\mathrm{SR}^2 \tag{B.160}$$

which is the version derived in Lo (2002).

B.13 Estimator Standard Errors

Exact standard errors of estimators exist only in very special cases such as the sample mean, least–squares regression estimators and the unbiased sample variance estimator, and under an assumption of normality. However, one may use an estimator's asymptotic variance to compute an approximate standard error (SE) of the estimator, and this is one of the most important uses of estimator asymptotic variance expressions. We discuss this use in the context of an estimator $\hat{\theta}_n$ of a true but unknown scalar parameter θ, based on an i.i.d. sample of size n,

B.13.1 Standard Errors Based on Asymptotic Variance Formulas

In order to compute a SE for an estimator $\hat{\theta}_n$, the first step is to determine the formula for the asymptotic variance $V(\theta) = V\left(\theta; \hat{\theta}_n\right)$. If the value of the parameter value θ were known, then the formula for an approximate SE of an estimator $\hat{\theta}_n$ would be:

[28] Mertens (2002)derived this result in terms of the coefficient of kurtosis γ_4, rather than the coefficient of excess kurtosis K_4, and so his formula with the SR^2 terms combined, has $K_4 + 2$ replaced with $\gamma_3 - 1$.

$$SE\left(\hat{\theta}_n;\theta\right) \approx \frac{1}{\sqrt{n}} V^{1/2}(\theta). \tag{B.161}$$

But since the value of θ is not known, we can replace the unknown θ with its estimate $\hat{\theta}_n$ to obtain the following estimate of the approximate SE:

$$SE\left(\hat{\theta}_n\right) \approx \frac{1}{\sqrt{n}} V^{1/2}(\hat{\theta}_n). \tag{B.162}$$

Comment: We typically use a "hat" symbol over parameter estimates, and if we followed that convention we would use $\widehat{SE}\left(\hat{\theta}_n\right)$. But for sake of simplicity of notation we do not do so, and it will be understood that $SE\left(\hat{\theta}_n\right)$ is an estimate of a standard error, that we refer to simply as the computed "standard error".

Example 1.1. Volatility Estimator Standard Error

In the case of the volatility estimator $\hat{\sigma}_{vol,n}$ the SE is:

$$SE\left(\hat{\sigma}_{vol,n}\right) = \frac{1}{\sqrt{n}} \left(\frac{\hat{\mu}_{4,n} - \hat{\sigma}^4_{vol,n}}{4\hat{\sigma}^2_{vol,n}} \right)^{1/2}, \quad \text{where} \quad \hat{\mu}_{4,n} = \frac{1}{n} \sum_{t=1}^{n} \left(r_t - \bar{r}\right)^4. \tag{B.163}$$

B.14 Fisher Information and MLE Asymptotic Optimality

Suppose that $(r_1, r_2, \cdots, r_n)$ are i.i.d. random variables $(r_1, r_2, \cdots, r_n)$ with pdf $f(\mathbf{r};\theta)$. Then with $l(\theta;\mathbf{r}) = \ln f(\mathbf{r};\theta)$ the natural log of the pdf, the *Fisher Information* for estimating the parameter θ is defined as:

$$I(\theta) = \mathrm{E}\left[\left(\frac{\partial l(\theta;r)}{\partial\theta}\right)^2\right]. \tag{B.164}$$

If z$\hat{\theta}_{mle,n} = \hat{\theta}_{mle,n}(r_1, r_2, \cdots, r_n)$ is a sequence of consistent MLEs of θ, then it can be shown that $\hat{\theta}_{mle,n}$ is asymptotically normal with asymptotic variance $I^{-1}(\theta)$:

$$\sqrt{n}\left(\hat{\theta}_{mle,n} - \theta\right) \xrightarrow{d} N\left(0, I^{-1}(\theta)\right). \tag{B.165}$$

Furthermore, any other consistent estimator, say $\hat{\theta}_n$, has an asymptotic variance $V\left(\hat{\theta}_n;\theta\right)$ that satisfies the information inequality

$$V\left(\hat{\theta}_n;\theta\right) \geqslant I^{-1}(\theta). \tag{B.166}$$

The above results reflect the asymptotic optimality of an MLE $\hat{\theta}_{mle,n}$.

B.15 Generalized Inverse of Distribution Functions

For any distribution function that has one or more "flat" spots, such as $F(x) = c$ for some constant c on some interval $x \in (a,b)$ with $a < b$, the inverse function $F^{-1}(c)$ does not exist.[29] An empirical

[29] The notation (a,b) means all x such that $a < x < b$, and the notation $(a,b]$ means all numbers x such that $a < x \leqslant b$.

distribution function $F_n(r)$ is an extreme version of this in that it is a pure step function consisting only of flat spots with values $p_i = i/n, i = 1, 2, \cdots, n$. One can handle the presence of flat spots in a distribution function through use of the following generalized inverse function:

$$F^{\leftarrow}(p) = inf\{r : p \leqslant F(r)\}, 0 < p \leqslant 1. \tag{B.167}$$

and takes on each of those values for many values of the argument r.

Application to Empirical Distribution Functions

For the EDF F_n of a sample $r_1, r_2, \cdots, r_n$ that has no ties, the ordered data satisfy the strict inequalities $r_{(1)} < r_{(2)} < \cdots < r_{(n)}$, and the generalized inverse of F_n is:

$$F_n^{\leftarrow}(p) = r_{(i)}, p \in ((i-1)/n, i/n], i = 1, 2, \cdots, n. \tag{B.168}$$

So for example, $F_4^{-1}(p) = r_{(2)} \forall p \in (1/4, 1/2]$, which may be written as $F_4^{-1}(p) = r_{(2)} \forall 4p \in (1, 2]$. What this reveals is that the γ–quantile of an EDF for a returns sample of size n is:

$$q_{F_n}(p) = F_n^{\leftarrow}(p) = r_{(\lceil np \rceil)}.$$

where $\lceil x \rceil$ is defined as the smallest integer greater than or equal to x. Thus, for example, when $n = 50$ and $p = 0.01$, we have $q_{F_{50}}(0.01) = r_{(\lceil 0.5 \rceil)} = r_{(1)}$.

Additional Comments

With $n = 50$ and $p = 0.02$, the height of the first flat spot of $F_n(r)$ is .02 for all $r \in [r_{(1)}, r_{(2)})$, and in this case it is not so clear that $r_{(1)}$ should be taken as the empirical quantile. To deal with this kind of issue, more mathematical treatments of risk consider also the "right" generalized inverse

$$F^{\rightarrow}(p) = inf\{r : p < F(r)\}, 0 < p \leqslant 1 \tag{B.169}$$

as well as the "left" generalized inverse B.167. For example Acerbi and Tasche (2002) take the empirical quantile, and hence empirical VaR, to be based on the right generalized inverse. In this case the empirical quantile will be $r_{(\lceil np \rceil)}$ with the ceiling function $\lceil x \rceil$ now defined as the smallest integer greater than x.

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